

Jev and Beyond: A Survey of Calibration-Aware Decision-Making with Language Models

GAYTRI JENA*, UC Berkeley, USA

KAPIL WANASKAR*, San Jose State University, USA

A language model that is to act, and not only to converse, must state how likely each of its answers is to be right, and software must be able to act on that number: answer, abstain, or hand the case off. Calibration research long treated this confidence as something estimated after the fact, from token probabilities, a post-hoc map, a prompted self-report or repeated sampling. Mostly since 2024, a second line of work puts calibration into training itself, as a supervised objective or as a reward. This survey maps both, with the training stage as its core. We collect 408 primary works and place each by the stage at which calibration enters the decision pipeline: the model’s own read-off, a post-hoc map, prompt elicitation, extra inference, or training, plus the works that act on a confidence or fix the output contract. The 80 training-stage works, 75 of them from 2024 to 2026, are organised by what the training signal scores, in five families: supervised calibration objectives, reward-model calibration, calibration rewards on the stated confidence, abstention and refusal training, and process- and meaning-level confidence rewards. Every work is also classified by the output contract its confidence is attached to. A typed decision, a probability for each declared option, is common at the read-off and post-hoc stages, but only 7 of the 80 training works train one; nearly all the rest train a stated score or a free-text hedge or refusal. Of the 39 closest prior surveys, none organises calibration training across domains by what the training signal scores, and none crosses that typology with the output contract. We close with seven measurable quantities, each tied to a public testbed, and report the first of them, the calibration of the decision itself, for Jev, a hosted typed-decision model, and 67 open decision models on a public leaderboard.

CCS Concepts: • **Computing methodologies** → **Natural language processing; Reinforcement learning; Mathematics of computing** → *Probability and statistics*.

Additional Key Words and Phrases: calibration, uncertainty, language models, reinforcement learning, proper scoring rules, abstention, conformal prediction, structured output, decision-making, survey

1 INTRODUCTION

A language model that automates a step in a program has to do more than produce a good answer. It has to say how likely that answer is to be right, in a form that the program can act on: answer when the probability is high, and abstain or hand the case to another model or a person when it is not. The rule is only as good as the number it reads. A confidence is *calibrated* when, among the cases to which the model gives probability p , a fraction p is correct [Guo et al. 2017]; a threshold policy built on a miscalibrated confidence answers when it should not, or escalates cases it could have settled.

Public typed-decision models. Tables 1 and 2 record two public views of the typed-decision models available on 2026-09-27; every row links to the model, so a reader can try one directly. Table 1 classifies the 268 repositories that the Hugging Face model search for “jev” returns: 76 only share the name, 4 could not be verified, and 188 were classified from their model cards and file lists; 137 contain newly trained weights, and general-purpose models with their copies account for 87% of downloads. A name search is a poor census of decision models: of the 51 open systems with released weights on the leaderboard of Table 2, it returns 4. Table 2 lists the top of that leaderboard, the Jev Decision Index [multimodalart 2026] (release 0.2.1), which scores 68 systems on skill over 38 decision benchmarks and on the calibration of the chosen option. Jev, a hosted model from

*The authors contributed to this work independently of their roles and employment.

Table 1. **Repositories returned by the Hugging Face model search for “jev”, by application** (snapshot of 2026-09-27). *Jev* is a hosted model from TypeSafe AI, released on 2026-09-15 [Almeida 2026], that returns a probability for each declared option (a typed decision, Table A1). Of the 268 repositories the search returns, 76 only share the name (75 predate Jev) and 4 could not be verified (empty, without a card, or gated with no public card metadata); the remaining 188 are classified here from their model cards and file lists (for a gated repository, from its public card metadata), with an application category taking precedence over a language one. Downloads are over the last 30 days. 137 repositories contain newly trained weights; the 38 copies, ports and packages add none (the 6 mirrors contain their source’s weight files byte for byte). General-purpose models and the copies of them account for 87% of the 41,653 downloads in the table. A name search misses most decision models: of the 51 open systems with released weights on the Jev Decision Index (Table 2), it returns 4.

Category	Repositories	Downloads	Examples
General-purpose typed-decision model	55	19,655	<i>JevK5</i> , <i>Open-Jev-9B</i> , <i>open-jev-deberta</i> , <i>Tiny-Jev</i> , <i>Jev-Style v3</i>
Research and ablation checkpoints	45	212	<i>jevify</i> (31-checkpoint readout series), <i>next-jev</i> (latent-workspace study), <i>mmbu-jev</i> (closed-set vs free-text)
Copy, port or package (no new training)	38	18,640	<i>JEV-9B-GGUF</i> (GGUF), <i>Jev-Omni-MLX-4bit</i> (MLX), <i>open-jev-deberta-ONNX</i> (ONNX), <i>JevK5-FP8</i> (FP8), <i>junetask/Open-Jev-9B</i> (a mirror)
No newly trained model (recipe, runtime, code or data)	11	0	<i>bonzi</i> (recipes), <i>JEV-CPU</i> (code), <i>jev-distill-corpus</i> (dataset)
Agents, tools and safety	10	1,005	<i>mini-Jev</i> (tool routing), <i>Mini-Jev shell-safety</i> , <i>Jev-Gate</i> (memory gating), <i>jev-ar</i> (Gulf-Arabic intent routing)
Domain application	10	152	<i>jev-legal</i> (Chinese legal), <i>jev-decider</i> (bank transactions), <i>jev-nyotti</i> (BTC trading), <i>Serigy-Jev</i> (PT-BR fake news), <i>Nano-Jev</i> (RAG)
Language-specific	7	1,010	<i>modernbert-ja-jev</i> (Japanese), <i>bielik-minitron-jev</i> (Polish), <i>Jev-RU</i> (Russian, Ukrainian, Belarusian)
Multimodal or vision	5	594	<i>Jev-Omni</i> (text, image, audio, video), <i>Jev-Vision</i> , <i>Jev-Spatial</i>
Games and control	5	156	<i>JevFlash</i> (Doom), <i>gemma-3-270m-jev</i> (ViZDoom), <i>JevAI</i> (Hearts of Iron IV mod)
Embeddings trained on typed-decision questions	2	229	<i>JevEmbed-KaLM</i> , <i>JevEmbed-Qwen3</i>
Total	188	41,653	

TypeSafe AI released on 15 September 2026 [Almeida 2026], returns a probability for each declared option; it is not one of the 408 works in our corpus, and its base model is not disclosed. Neither view is peer reviewed, and neither is used to rank the training families of Section 3.1.

For most of the literature, the confidence of a language model is estimated *after* the model has been trained. It is read off the model’s own token or sequence probabilities, which are well calibrated on multiple-choice formats for large pretrained models [Kadavath et al. 2022] but lose calibration after post-training [OpenAI 2023]; it is rescaled by a post-hoc map fitted on held-out data [Guo et al. 2017]; it is requested in words or numbers through the prompt [Tian et al. 2023], where stated confidences cluster near the top of the scale [Xiong et al. 2023]; or it is computed from extra inference, such as the agreement of sampled answers or a conformal prediction set [Campos

Table 2. **Jev and open decision models on the Jev Decision Index**: ranks 1 to 15 of the 68 systems on the [Jev Decision Index](#) [multimodalart 2026] (release 0.2.1, data of 2026-09-27). Skill is chance-normalised (0 to 100) over the same 38-benchmark panel for every system, and unanswered requests count as wrong. ECE is measured on a one-in-6 sample of 32 benchmarks with a per-field right answer. The board reports point estimates without uncertainty intervals. Base model and method are as recorded by the board: full fine-tune, a low-rank adapter (LoRA), a trained head or adapter over a frozen model, inference only (prompting or decoding over unchanged weights) or a hosted API. Names link to the model’s Hugging Face repository, or to its code where no weights are released. Jev ranks 1 on skill; across the whole board, 17 of the 67 open systems have a lower ECE than Jev.

Rank	System	Base model	Method	Skill	ECE
1	Jev 1.13 (TypeSafe AI, hosted)	not disclosed	hosted API	57.91	0.074
2	Surogate Rune 26B-A4B v3 [bf16]	gemma-4-26B-A4B-it	full fine-tune	57.44	0.120
3	Decider chat · Gemma-4-31B	gemma-4-31B	inference only	57.33	0.047
4	AutoJev-27B	Qwen3.8-27B	full fine-tune	56.40	0.018
5	simple-jev · Qwen3.8-27B (featherless)	Qwen3.8-27B	inference only	55.74	0.113
6	Jebadiah 27B	Qwen3.8-27B	LoRA	54.67	0.014
7	reflex 27B [FP8 · wide choice]	Qwen3.8-27B	inference only	52.16	0.024
8	Decider chat · Qwen3.6-27B	Qwen3.6-27B	inference only	51.35	0.021
9	Winnow-12B [Q8_0]	gemma-4-12B	LoRA	50.02	0.168
10	JoshuaSP diffusion gemma (open-jev)	diffusiongemma-26B-A4B-it	inference only	49.47	0.216
11	Jevfire	Qwen3.8-27B	inference only	49.37	0.052
12	Decider 35B-A3B [NVFP4]	Qwen3.5-35B-A3B-Base	full fine-tune	47.11	0.023
13	JPT-9B	Qwen3.5-9B-Base	LoRA	46.89	0.077
14	Decision 1.0 Lux	Qwen3.5-9B-Base	head or adapter	43.49	0.076
15	Xor	Qwen3.6-35B-A3B	full fine-tune	41.48	0.015

et al. 2024; Farquhar et al. 2024]. Figure 1 shows examples of each stage, and of training, taken from the papers’ own figures.

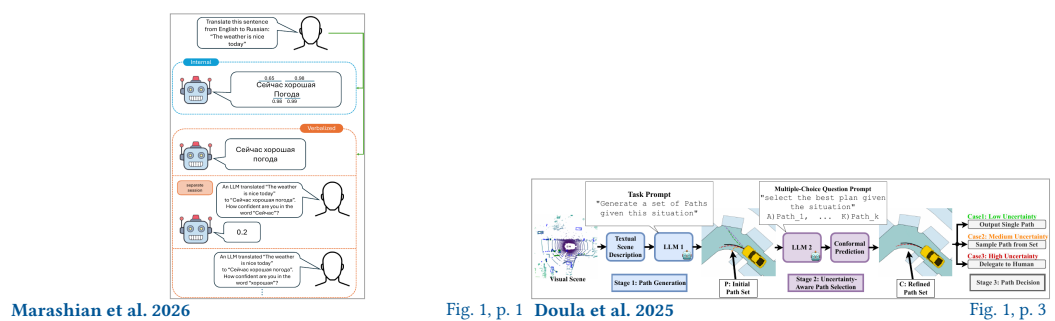
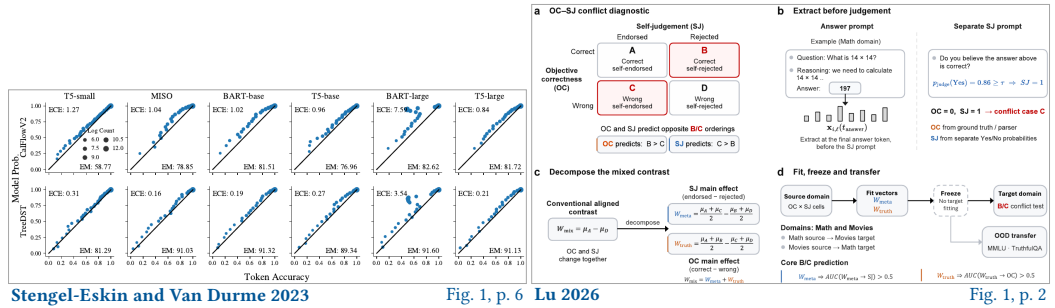
A second line of work, most of it from the last three years, puts calibration into training itself. The model is fitted to confidence targets or a calibration loss, the reward model in its training loop is calibrated, or a reinforcement-learning reward scores the stated confidence against the outcome with a proper scoring rule [Bani-Harouni et al. 2026b; Damani et al. 2025], rewards abstention when the model does not know [Wei et al. 2025], or scores confidence per reasoning step [Wang et al. 2026d]. Of the 80 works in our corpus that put a calibration term into the model’s own loss or reward, 75 appeared from 2024 to 2026 (94%). At the same time, models are offered that return, instead of free text, a probability for each option of a set declared in advance, so that the decision and its confidence come from one distribution; one such hosted model, Jev [Almeida 2026], and the open decision models listed with it on a community leaderboard [multimodalart 2026] are listed in Tables 1 and 2, and an independent evaluation of it is part of our corpus [Li et al. 2026c].

This survey maps calibration-aware decision-making with language models, with the training stage as its core. We place each of 408 works by the stage at which calibration enters the decision pipeline (Table A1), organise the 80 training-stage works by what the training signal scores, and classify every work by the *output contract* its confidence is attached to: a typed decision with a probability per declared option, a prediction set, constrained output, a single confidence score, or free text. The two classifications meet in one finding: of the 65 works that attach the confidence to a typed decision, most calibrate a classifier or a multiple-choice read-off after the fact, and only 7 of the 80 training works train one (Section 4).

Contributions. This survey offers the following.

Confidence estimated after the fact, and confidence trained into the model

Estimated after the fact: read-off, post-hoc maps, prompts, extra inference (S0 to S3)



Trained into the model: a calibration term in the training loss or reward (S4)

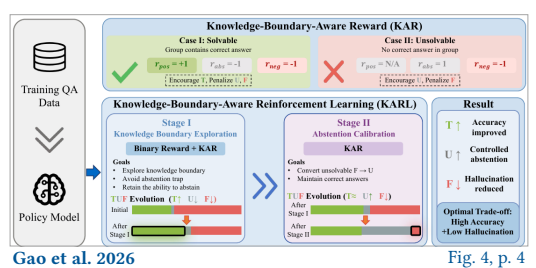
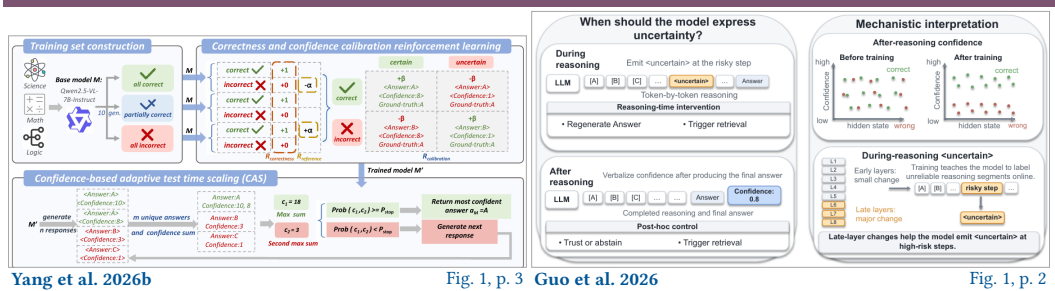


Fig. 1. Confidence estimated after the fact, and confidence trained into the model. Panels are drawn from Figures 8 and 9 under the same licence rule: one per stage for S0, S1, S2, S3, and one per training family for S4 (T1, T3, T4; T2, T5 not shown). Each label links to the paper; Table E1 gives the figure, page and licence.

- A **stage taxonomy** of 408 works (Figure 2 shows 77 of them), with operational placement rules (Table A1) and a per-work record of stage, output contract, access, guarantee, metric and action (Tables C1 and F1).
- A **typology of calibration training**: the 80 training-stage works in five families by what the training signal scores (Section 3.1, Table C1), and what each stage yields, costs and guarantees (Table 3).
- A **coverage comparison** with the 39 closest prior surveys (Table B1, Figure 3), and a profile of the corpus (Figures 5 to 9), every gallery panel traced to its source figure and licence (Table E1).
- An analysis of the **output contract and the decision loop** (Section 4, Figure 10, Table D1).
- A **measurable agenda**: seven quantities with named public testbeds (Table 4), with the calibration of the decision itself measured for the 68 systems of a public leaderboard (Figure 11).

1.1 Related surveys and the empty cell

Table B1 (Appendix B) compares the 39 closest prior surveys on seven topics: confidence estimation and calibration, calibration-aware training of any kind, reinforcement learning whose reward is a calibration or proper-scoring-rule objective, decisions over a predefined option set, typed per-option decision outputs, acting on confidence, and hallucination. Figure 3 places them on a timeline; 33 of the 39 appeared from 2024 onward.

Calibration and uncertainty are well surveyed, and the stage view itself, which organises methods by where the confidence comes from, is established [Li et al. 2025e; Shorinwa et al. 2025]; we use it as the frame for the background stages and credit it there. The training stage is the thinner cell. Only 3 surveys give full coverage to reinforcement learning with a calibration reward, and each is scoped to one slice: Li et al. [2025e] to the 2024 methods, Xu et al. [2026b] to forecasting, where it organises calibration training by reward and maps forecast forms to proper scores, and Lin et al. [2026] to agents, where it splits rewards by their target. Ulmer et al. [2025] labels papers by the register of the confidence, numeric or verbal, and by how it is elicited: by prompting, supervised fine-tuning or reinforcement learning. What none of the 49 surveys we identified does is organise calibration training *across domains* by what the training signal scores, or cross that typology with the output contract the confidence is trained into; no survey cites more than one of the 14 general-domain calibration-training works from 2025 and 2026 that anchor our core (marked in Table C1). Two recent non-survey works come closest: an application paper that compares reinforcement-learning training paradigms by their signal source for one domain [Barbosa 2026], and a position paper that groups ternary-reward abstention works, three of them in our core, in one row [Ravikumar 2026]; neither proposes a typology of training signals or an output contract. Typed per-option decision output is the emptiest column of Table B1: no survey covers it in full.

2 SURVEY METHODOLOGY

2.1 Search, deduplication and placement

The corpus was built from 7 web searches run on 2026-09-26, one per slice of the pipeline: the model’s read-off and post-hoc maps (S0 and S1), prompt elicitation (S2), sampling-based and conformal methods (two slices of S3), training (S4), the works that act on a confidence or fix the output contract (X), and independent evaluations of Jev, a hosted typed-decision model released during the search period. The query strings and the number of records screened were not recorded; Figure 4 marks those stages as not recorded rather than estimating them. Records were matched

Calibration-aware decision-making with language models		408 works, placed by the stage at which calibration enters
S0 read-off (25 works) Empirical studies of read-off calibration (14) Calibration Across Layers: Understanding... [Joshi et al. 2025] Calibration after alignment (5) Sanz-Guerrero: Large Language... [Sanz-Guerrero et al. 2026] Likelihood and logit read-off (5) My Answer is C: First-Token Probabilities... [Wang et al. 2024d] Self-evaluation and read-off calibration (1) Kadavath: Language Models (Mostly) Know... [Kadavath et al. 2022]	S3 extra inference (121 works) Conformal prediction sets (36) CommCP: Efficient Multi-Agent Coordination... [Zhang et al. 2026b] Sampling consistency (18) Too Consistent to Detect: A Study of... [Tan et al. 2025] Semantic entropy and meaning clustering (18) Sun: Efficient Hallucination Detection... [Sun et al. 2026a] Risk control and early exit (16) Conformal Thinking: Risk Control for... [Wang et al. 2026b] Conformal abstention, selection and cascades (6) Tayebati: Learning Conformal Abstention... [Tayebati et al. 2025] Conformal factuality and claim filtering (6) Rubin-Toles: Conformal Language... [Rubin-Toles et al. 2025] Benchmarks and empirical studies (5) Vashurin: Benchmarking Uncertainty... [Vashurin et al. 2025] Ensembles and cross-model disagreement (5) Hamidieh: Complementing Self-Consistency... [Hamidieh et al. 2026] Long-form and claim-level (5) IUQ: Interrogative Uncertainty Quantification... [Fan et al. 2026] Internal-state signals (3) Semantic Volume: Quantifying and Detecting... [Li et al. 2026c] Token-level aggregation (3) Word-Sequence Entropy: Towards Uncertainty... [Wang et al. 2025a]	
S1 post-hoc map (52 works) Probes and learned confidence heads (17) Lu: Diagnosing Correctness Probes under... [Lu 2026] Learned calibrators (9) Luo: Unlocking the Pre-Trained Model as a... [Luo et al. 2026] Prompt and context calibration (9) Gundem: Boosting In-Context Learning in... [Gundem et al. 2025] Scaling and binning (7) Luo: Your Pre-trained LLM is Secretly an... [Luo et al. 2025] Fixed maps over the read-off (6) The Silent Vote: Improving Zero-Shot LLM... [Badhe et al. 2026] Label-bias and option-prior debiasing (2) Beyond Performance: Quantifying and... [Reif and Schwartz 2024] Multicalibration and task-specific recalibration (2) Liu: On Calibration of LLM-based Guard Models... [Liu et al. 2024b]	X consumers and contract (71 works) Constrained decoding (19) Reddy: The Hidden Cost of Structured... [Reddy et al. 2026] Routing/cascade (13) Bouchard: Is Escalation Worth It? A... [Bouchard 2026] Structured-output benchmark (10) Singh: The Structured Output Benchmark: A... [Singh et al. 2026a] Human reliance on model output and stated confidence (7) Not All Uncertainty Is Equal: How... [Villavicencio et al. 2026] Adaptive compute (6) Yang: Dynamic Early Exit in Reasoning Models [Yang et al. 2025b] Deferral (4) L2D-Clinical: Learning to Defer for... [Kondadadi and Ortega 2026] Judge calibration (4) JEV-as-a-Judge: Accept When Confident... [Li et al. 2026c] Abstention (3) Srinivasan: Selective "Selective... [Srinivasan et al. 2024] Studies and benchmarks of abstention (3) Wang: Are LLM Decisions Faithful to Verbal... [Wang et al. 2026c] Confidence-weighted selection (1) Taubenfeld: Confidence Improves... [Taubenfeld et al. 2025] Typed classification (1) Look at the Text: Instruction-Tuned... [Wang et al. 2024c]	
S2 prompt elicitation (54 works) Verbalised numeric confidence (30) Yang: The Mirage of Calibrated Confidence... [Yang et al. 2026a] Linguistic hedges and epistemic markers (17) Liu: Can LLMs Use Linguistic Uncertainty... [Liu and Cohan 2026] Introspection, deliberation and task-specific elicitation (3) Reasoning about Uncertainty: Do Reasoning... [Mei et al. 2025] Stated confidence that triggers an action (2) Ferrer: When Can We Trust LLM Graders?... [Ferrer et al. 2026] Abstention prompting (1) I-CALM: Incentivizing Confidence-Aware... [Zong et al. 2026] Self-evaluation prompts (1) Tian: Overconfidence in LLM-as-a-Judge... [Tian et al. 2025]	Theory and analysis (5 works, not counted in the core) Blasiok: When Does Optimizing a Proper... [Blasiok et al. 2023] Kalai: Calibrated Language Models Must... [Kalai and Vempala 2024] Uncalibrated Reasoning: GRPO Induces... [Bereket and Leskovec 2025] Kalai: Why Language Models Hallucinate [Kalai et al. 2025] Nakkiran: Trained on Tokens, Calibrated... [Nakkiran et al. 2025]	
S4 training: the core (80 works), by what the training signal scores		
T1 supervised calibration objectives (23) Jiang: How Can We Know When Language Models... [Jiang et al. 2021] Chaudhry: Finetuning Language Models to... [Chaudhry et al. 2024] Kapoor: Large Language Models Must Be... [Kapoor et al. 2024] Niu: Functional-level Uncertainty... [Niu et al. 2024] Uncertainty Distillation: Teaching... [Hager et al. 2025] ConfTuner: Training Large Language Models to... [Li et al. 2025d] Direct Confidence Alignment: Aligning... [Zhang et al. 2025c] Know When You're Wrong: Aligning Confidence... [Xie et al. 2026] T2 reward-model calibration (5) Leng: Taming Overconfidence in LLMs: Reward... [Leng et al. 2025] Lou: Uncertainty-aware Reward Model: Teaching... [Lou et al. 2024] Know What You Don't Know: Uncertainty... [Park et al. 2025a] Ma: Distributional Process Reward Models... [Ma et al. 2026a] Pan: Uncertainty-Aware Reward Modeling for... [Pan et al. 2026a] T5 process- and meaning-level confidence rewards (4) An: Teaching LLMs to Abstain via Fine-Grained... [An and Xu 2025] MMBoundary: Advancing MLLM Knowledge Boundary... [He et al. 2025] Wang: Process Supervision of Confidence... [Wang et al. 2026d] Yu: Calibrating LLMs with Semantic-level Reward [Yu et al. 2026b]	T3 calibration rewards on the stated confidence (31) Band: Linguistic Calibration of Long-Form... [Band et al. 2024] Rewarding Doubt: A Reinforcement... [Bani-Harouni et al. 2026b] Niekerk: Post-Training Large Language... [van Niekerk et al. 2025] Zou: Thinking, Faithful and Stable... [Zou et al. 2025] ORCE: Order-Aware Alignment of Verbalized... [Li et al. 2026a] Singh: Verifiable Rewards for Calibrated... [Singh et al. 2026b] DualStake: Dual-Path Confidence Calibration in... [Xu et al. 2026a] Closing the Reflection Gap: A Free... [Zhu 2026] T4 abstention and refusal training (17) Yang: Alignment for Honesty [Yang et al. 2024b] Chen: Teaching Large Language Models to... [Chen et al. 2025a] Know the Unknown: An Uncertainty-Sensitive... [Li et al. 2025c] Xu: Rejection Improves Reliability: Training... [Xu et al. 2024b] Honesty over Accuracy: Trustworthy... [Mohamadi et al. 2025] TruthRL: Incentivizing Truthful LLMs via... [Wei et al. 2025] TIAR: Trajectory-Informed Advantage... [Pan et al. 2026b] Abstain-R1: Calibrated Abstention and... [Zhai et al. 2026]	

Fig. 2. **A taxonomy of calibration-aware decision-making with language models** (77 of the 408 works in Tables C1 and F1). Blocks are the stage at which calibration enters (Table A1); X holds the works that consume a confidence or fix the output contract. Every method family at S0 to S3 and X shows its size and its most recent work. The core, S4 (80 works), is grouped by what the training signal scores and lists up to 8 works per family, evenly spaced over the family's works in year order. The 5 theory and analysis works are listed in full and are not counted in the core. Each entry gives a trimmed form of the paper's own title; a citation may show a later venue year than the first-version year counted in Figures 6 and 7.

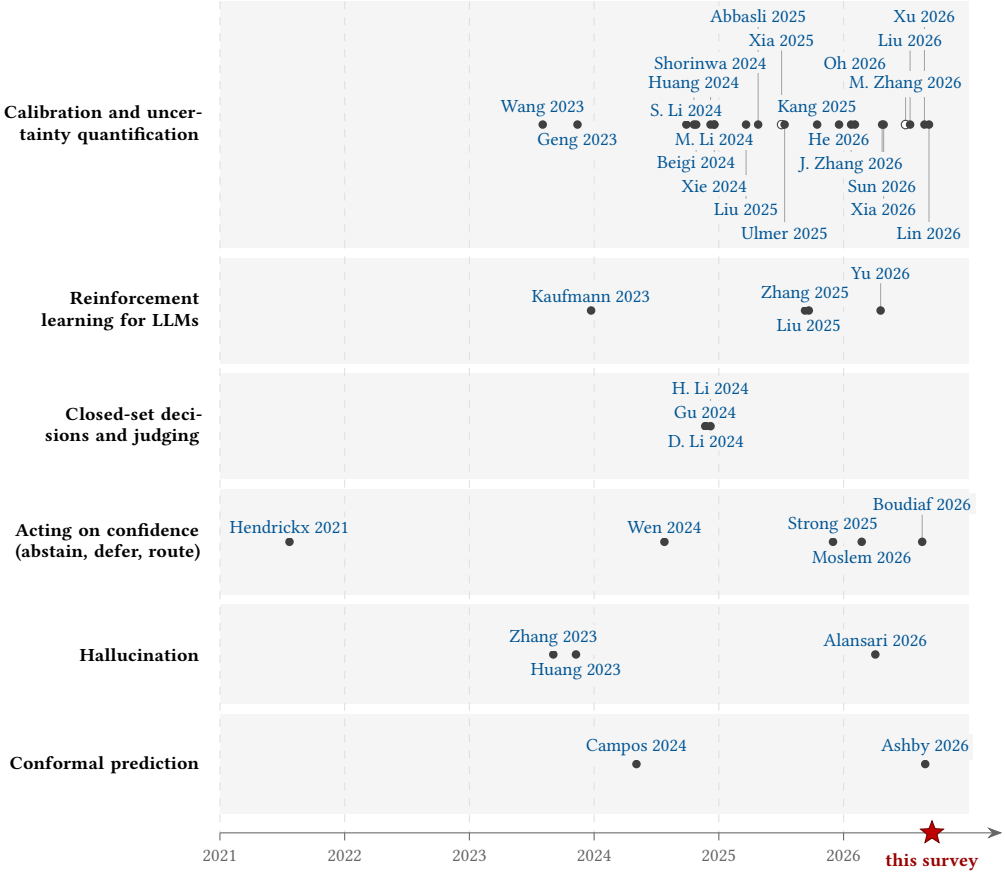


Fig. 3. **The 39 closest prior surveys on a timeline, one lane per topic group** (Table B1). Each mark sits at the survey’s first public version (the arXiv first version where one exists); a hollow mark means only the year is known, placed mid-year. Each label links to the survey. 33 of the 39 surveys appeared from 2024 onward. None fully covers typed decision outputs (1 partial: Lin 2026), and each full treatment of calibration-reward training is scoped to one slice (Table B1). The star on the time axis marks this survey.

on arXiv identifier, DOI and normalised title: 22 duplicates were removed and 11 records were matched to references already cited and kept, leaving 408 works.

Placement. A work is in the corpus when its main contribution meets one of the stage definitions of Table A1: it produces, maps, elicits, samples or trains the confidence of a language model (S0 to S4), analyses how training changes calibration, or acts on a confidence or fixes the output contract (X). Each work is placed by what its main contribution changes, not by the task it is evaluated on, and each training-stage work was assigned to one family by reading it. A work is *core* if and only if a calibration term enters the model’s own training loss or reward; the 5 theory and analysis works are listed with the core but not counted in it, and the 323 works at the other stages form the landscape (Appendix F). The per-work fields of Tables C1 and F1 (output contract, access, samples, guarantee, trained, metric, task and action) were recorded for each paper. The fields were checked against the papers in several review rounds and corrected where they disagreed, and a value that cannot be determined from the paper is marked as such rather than guessed.

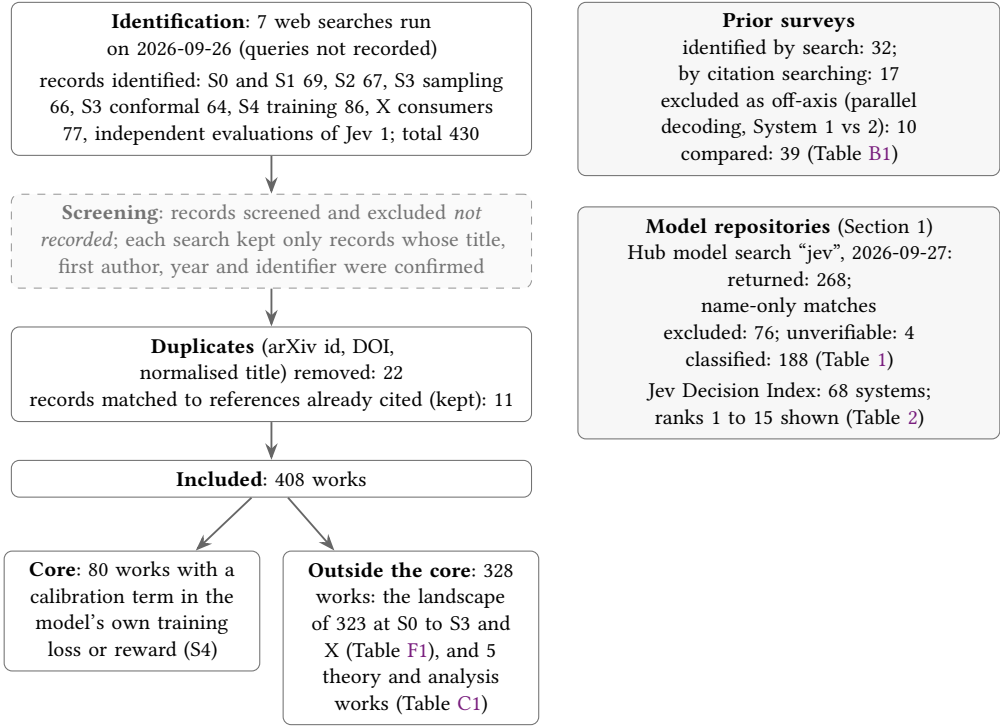


Fig. 4. **Corpus construction flow**, laid out after PRISMA 2020. The main column counts the primary works; the side boxes count the prior surveys and, for Tables 1 and 2, the model repositories and the leaderboard. Screening counts were not recorded and are marked rather than estimated. A work is core if and only if a calibration term enters the model's own training loss or reward (Table A1); theory and analyses of such training are listed with the core in Table C1 but not counted in it.

Prior surveys. 32 prior surveys were identified by search and the rest by backward citation searching from the surveys found, 49 in all. 10 of them, on parallel and speculative decoding [Gwak et al. 2026; Hu et al. 2025; Li et al. 2025a; Ryu and Kim 2024; Xia et al. 2024b; Xiao et al. 2023; Yu et al. 2025a; Zhou et al. 2024c] and on fast versus slow reasoning [Li et al. 2025f; Sui et al. 2025], are off the axis of this survey and are set aside; the remaining 39 are compared in Table B1. The marks for reinforcement learning with a calibration reward were checked in the full text or the full reference list of every survey.

2.2 The corpus at a glance

Figure 5 summarises the corpus and Figure 6 profiles it. Extra inference (S3) is the largest stage, with 121 works, most of them conformal prediction sets, risk control and sampling-based uncertainty; training (S4) follows with 80, and the consumers of a confidence (X) with 71. Most methods need the model's logits or weights (221 white-box works against 126 that work from text alone). Figure 7 shows how recent the training stage is: 75 of its 80 works appeared from 2024 to 2026 (94%), against 166 of the 252 works at the other stages over the same years (66%).

Figures 8 and 9 show, stage by stage, how the papers themselves depict their methods and their results. Panels are chosen by a fixed rule and reproduced only from papers whose licence permits reuse; the rule, and each panel's source figure, page and licence, are in Table E1 (Appendix E).

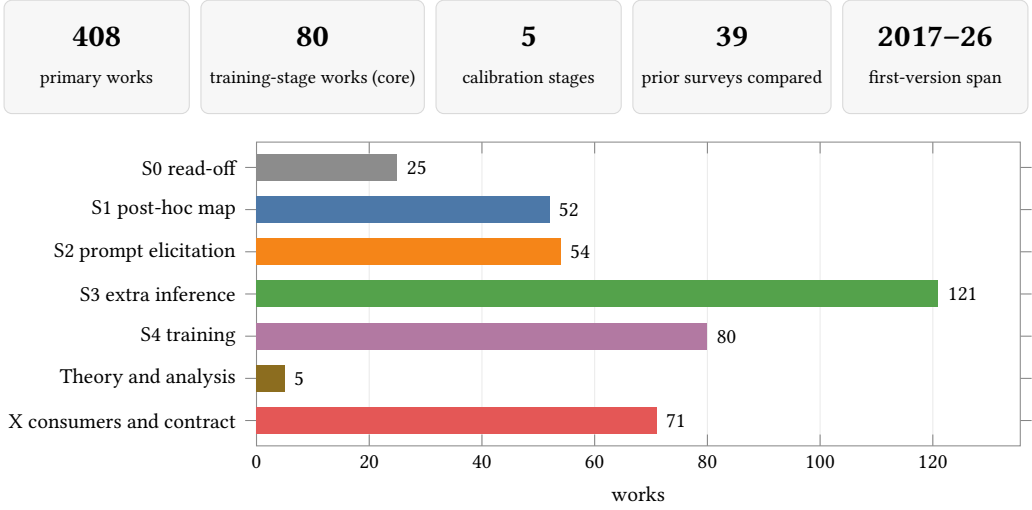


Fig. 5. **The corpus at a glance.** 408 primary works (451 references cited across the figures and tables), placed by the stage at which calibration enters (Table A1); the X group holds works that consume the confidence or fix the output contract, and the theory group holds analyses of training-stage calibration. 39 prior surveys are compared in Table B1.

3 A TAXONOMY OF WHERE CALIBRATION ENTERS

The taxonomy of Figure 2 places each work by *where the calibration signal enters the decision pipeline*. Table A1 (Appendix A) gives the operational definition of each stage as a pair: what counts, and the nearest case that does not. Five stages change how the confidence is produced: the model’s own read-off (S0), a post-hoc map (S1), prompt elicitation (S2), extra inference (S3) and training (S4). The consumers (X) use a confidence without changing how it is produced, and the theory and analysis works explain when training yields calibration. The last three rows of the table define the properties recorded for every work: the output contract, the action a confidence triggers when it is low, and whether a distribution-free guarantee is stated.

3.1 The core: calibration in the training loss or reward (S4)

The 80 core works share one property: a calibration term enters the model’s own loss or reward, so that the model itself, and not a map or a procedure around it, is changed towards calibration. We group them by *what the training signal scores*. Table C1 (Appendix C) lists every core work with its family, output contract, action, guarantee and metric.

T1: supervised calibration objectives (23 works). The model is fitted to confidence targets or a calibration loss, without a reward. The oldest works regularise fine-tuning for calibration in and out of distribution [Kong et al. 2020] or teach a model to state a calibrated numeric or verbal confidence [Lin et al. 2022]. Later work tunes on graded sets of correct and incorrect answers [Kapoor et al. 2024], distils sampling-based uncertainty into a single-pass confidence [Hager et al. 2025; Huang et al. 2025a], and trains the stated confidence with a proper scoring rule as the loss [Li et al. 2025d]. One line targets calibration lost elsewhere in training: after preference alignment [Xiao et al. 2025] or after reinforcement learning [Xie et al. 2026].

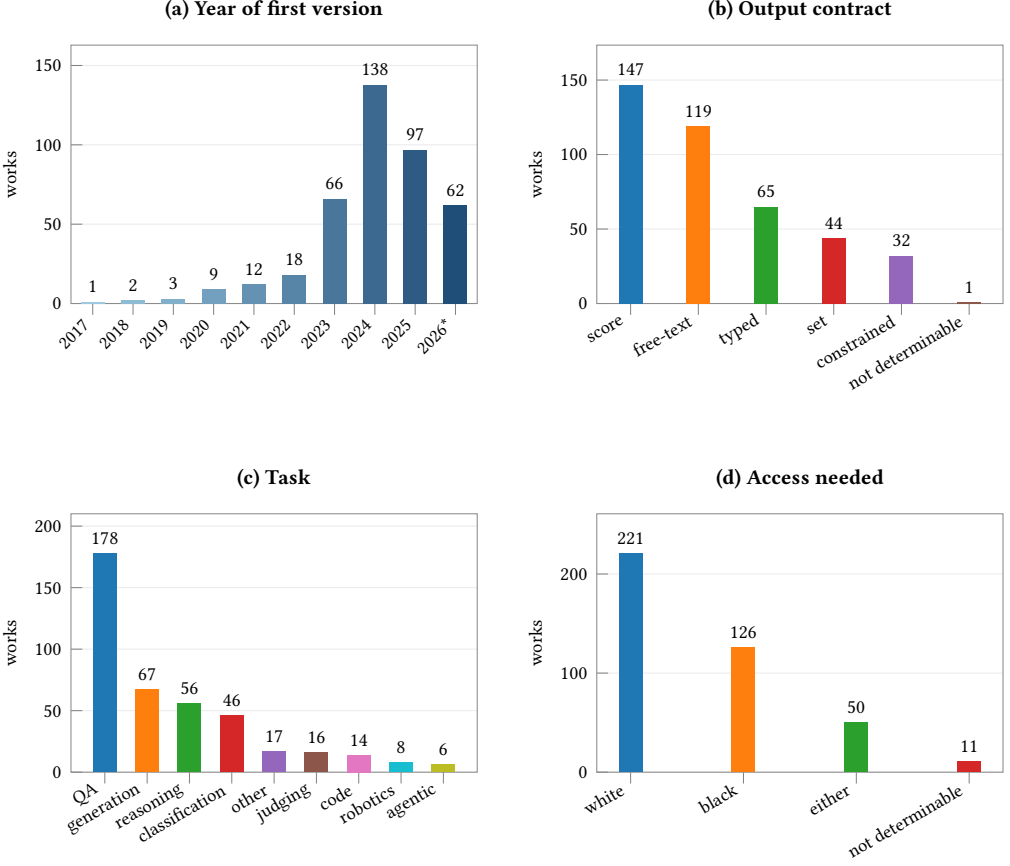


Fig. 6. **Profile of the 408 corpus works** (Tables C1 and F1). (a) Year of first public version (*2026 is partial); (b) the output contract the confidence is attached to (the five contracts are defined in the caption of Table F1); (c) task; (d) the access a method needs: white-box (logits or weights), black-box (text only) or either. “not determinable” marks a value that could not be determined from the paper.

T2: reward-model calibration (5 works). The calibrated model is a reward model, usually the one in the training loop. Calibrated reward modelling counters the overconfidence that preference optimisation induces [Leng et al. 2025]; other works expose the reward model’s own uncertainty as a variance [Lou et al. 2024] or calibrate process reward models by quantile regression, here to set inference compute per instance [Park et al. 2025a]. One reward-model method adds conformal bounds [Pan et al. 2026a]; it is the only core work that states a distribution-free guarantee.

T3: calibration rewards on the stated confidence (31 works). A reward or preference scores the confidence the model states against the outcome. This is the largest family and, with T5, the newest: 27 of its 31 works appeared in 2025 or 2026, 17 of them in 2026. The rewards include the logarithmic scoring rule [Bani-Harouni et al. 2026b], a Brier term added to the correctness reward [Damani et al. 2025], and rewards that let a simulated reader make calibrated forecasts [Band et al. 2024]. Later work asks which confidence rewards cannot be hacked [Tan et al. 2026], compares proper scoring rules as terminal rewards for forecasters [Turtel et al. 2026], separates the reasoning and calibration objectives to avoid gradient conflict [Ma et al. 2026b], and adjusts the probabilities of

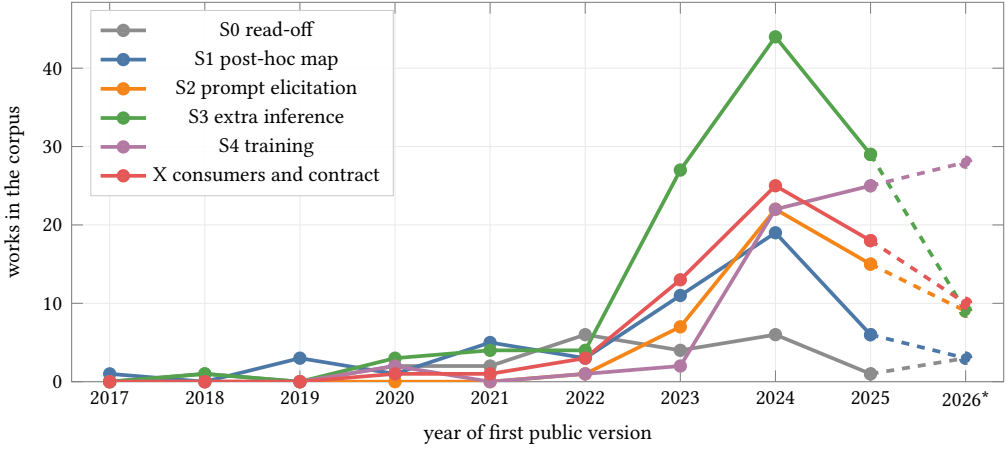


Fig. 7. **Works per stage per year** (Tables C1 and F1). 75 of the 80 training-stage (S4) works appeared in 2024 to 2026 (94%), against 166 of the 252 works at stages S0 to S3 (66%) over the same years. *2026 covers January to September, drawn dashed. The 5 theory and analysis works are not plotted.

decision tokens directly [Yaldiz et al. 2026]. Ranking-based confidence rewards [Wang et al. 2026a] target discrimination as well as calibration.

T4: abstention and refusal training (17 works). The signal targets the action: the model is trained, by supervision or by reward, to abstain when it does not know. Early work builds refusal-aware fine-tuning data from the questions the model gets wrong [Yang et al. 2024b; Zhang et al. 2023a]. Reinforcement-learning variants use a ternary reward that scores a correct answer above an abstention and an abstention above an error [Mohamadi et al. 2025; Wei et al. 2025], add clarification after a refusal [Zhai et al. 2026], or derive the reward from the model’s knowledge boundary [Gao et al. 2026]. The family also carries a warning: error-penalised abstention rewards can collapse a KL-anchored policy to refusing everything [Che et al. 2026].

T5: process- and meaning-level confidence rewards (4 works). The confidence reward is given per reasoning step [He et al. 2025; Wang et al. 2026d] or over answers that are equivalent in meaning [An and Xu 2025; Yu et al. 2026b], rather than once per final answer.

Theory and analysis (5 works). These works explain when training yields calibration and are not counted in the core. Local optimality of a proper loss implies smooth calibration [Błasiok et al. 2023]; a calibrated language model must hallucinate arbitrary facts at a rate close to the fraction of such facts seen exactly once in training [Kalai and Vempala 2024], and binary-graded training and evaluation reward guessing [Kalai et al. 2025]. Calibration that emerges from next-token training can break under reinforcement-learning tuning [Nakkiran et al. 2025], and the group standard-deviation normalisation of GRPO causes overconfidence on stochastic outcomes [Bereket and Leskovec 2025].

Families crossed with the output contract. The families differ in what they score; they also differ in what form the trained confidence takes. Of the 80 core works, 49 train a single confidence score, 23 a free-text answer with a hedge or refusal, and 7 the probability of a typed decision, all of which report its calibration; 1 has another or an undetermined contract. The typed decision appears only in T1 and T3, and abstention training (T4) is almost entirely free text:

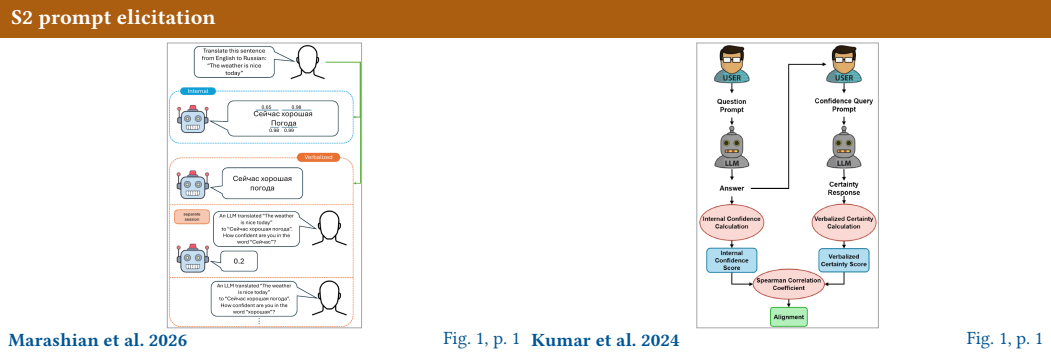
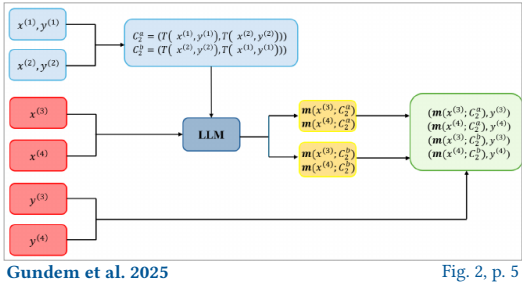
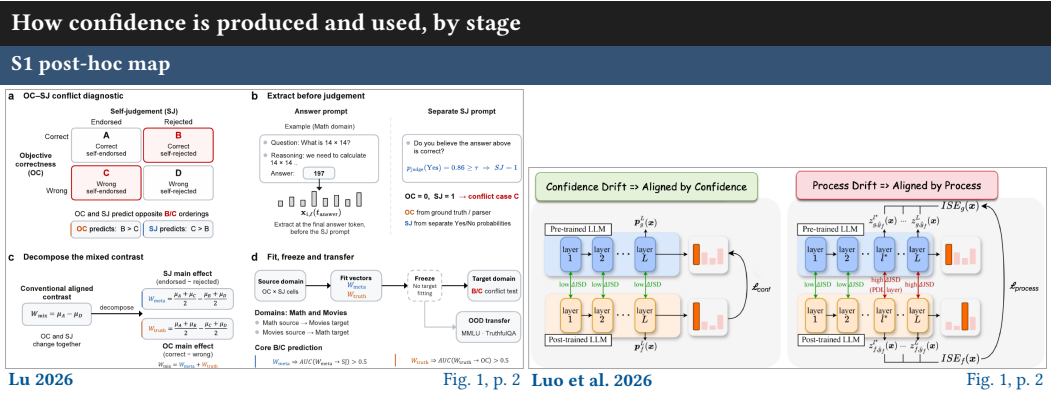


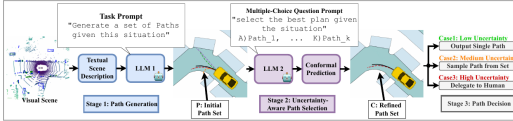
Fig. 8. **How confidence is produced and used, by stage.** Each panel shows a whole figure of the cited paper, reproduced under its CC BY licence; its label links to the paper. The selection rule and each panel's figure, page and licence are in Table E1. S0 read-off has no panel: no reusable figure passed the check (Table E1).

Family	Score	Free text	Typed	Other	All
T1 supervised calibration objectives	15	2	6	0	23
T2 reward-model calibration	4	0	0	1	5
T3 calibration rewards on the stated confidence	27	3	1	0	31
T4 abstention and refusal training	2	15	0	0	17
T5 process- and meaning-level confidence rewards	1	3	0	0	4
All core works	49	23	7	1	80

Other: a prediction set, or a contract that could not be determined from the paper.

How confidence is produced and used, by stage (continued)

S3 extra inference



Doula et al. 2025

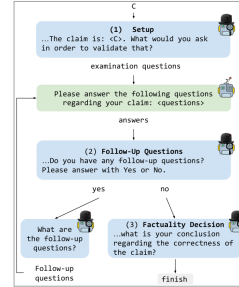
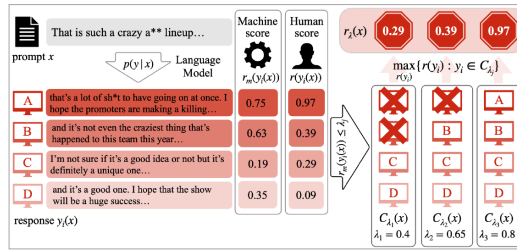


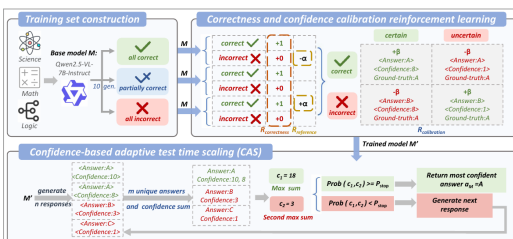
Fig. 2, p. 3



Chen et al. 2025b



S4 training



Yang et al. 2026b

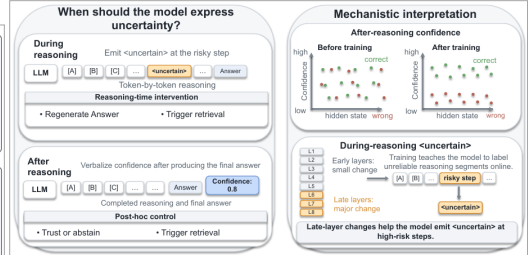


Fig. 1, p. 2

Fig. 8. How confidence is produced and used, by stage (continued).

The action recorded for a core work is most often none (45 works): the trained confidence is reported with the answer, and the rule that would act on it is left to the user. Abstention is the exception, because the T4 family trains the action itself (21 core works abstain).

3.2 The background stages and the consumers

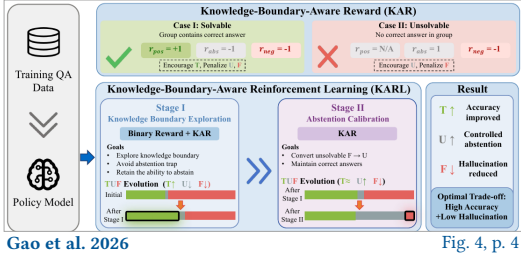
The stages outside the core are organised in the existing surveys [Li et al. 2025e; Shorinwa et al. 2025]. Table 3 compares them with the core on what each yields, costs, guarantees and gets wrong, and Table F1 (Appendix F) lists all 323 works.

The read-off (S0, 25 works) needs token probabilities; post-hoc maps (S1, 52 works) add temperature scaling, binning [Guo et al. 2017] and probes on hidden states [Azaria and Mitchell 2023]; prompt elicitation (S2, 54 works) works from text alone [Tian et al. 2023]. Extra inference (S3, 121 works) samples several answers and measures their agreement [Farquhar et al. 2024; Wang et al.

Table 3. **What each stage yields, costs, guarantees and gets wrong.** The second column is the sense in which the literature uses “confidence” at that stage. A cited phrase is a finding of the cited work; an uncited phrase follows from the definition of the stage (Table A1). Only training (S4) changes the model itself; its families are compared in Table C1.

Stage	Confidence it yields	Cost: data; inference	Guarantee	Black-box use	Typical failure	Fit for a typed decision
S0 read-off	the model’s own probability for an option or a true/false judgement [Kadavath et al. 2022]	none; none	none	no: needs token probabilities	calibration degrades after post-training [OpenAI 2023]	natural when the options are enumerated and scored [Kadavath et al. 2022]
S1 post-hoc map	a rescaled, binned or pooled probability [Guo et al. 2017], a probe’s probability from the hidden states [Azaria and Mitchell 2023], or an auxiliary model’s estimate from the input and output text [Ulmer et al. 2024a]	held-out data (labels usually; none for a fixed map); negligible	binning: distribution-free [Gupta and Ramdas 2021]; scaling: none	partly: a map needs the model’s scores and a probe its hidden states; an auxiliary model on the text needs neither	a map fitted for one task must be refitted, or predicted by a learned model, for a new one [Shen et al. 2024]	good for a fixed label set [Guo et al. 2017]
S2 prompt elicitation	a confidence stated in words or numbers [Tian et al. 2023]	none; none if asked with the answer, one more call if asked afterwards	none	yes	stated confidences cluster near the top of the scale [Xiong et al. 2023]	the confidence arrives as text and must be parsed
S3 sampling	agreement or semantic entropy across sampled answers [Farquhar et al. 2024; Wang et al. 2023b]	none; several sampled answers per query	none	yes	consistent but wrong answers look confident [Tan et al. 2025]	defined over free text; per-option use needs clustering
S3 conformal and risk control	a prediction set of options [Campos et al. 2024], or a threshold with bounded risk, such as an early exit [Schuster et al. 2022]	a calibration set; a threshold over one pass, or over several sampled answers for sets of generations	coverage $\geq 1 - \alpha$ [Campos et al. 2024], or a risk bound [Angelopoulos et al. 2025]	either	the guarantee lapses when exchangeability fails [Ulmer et al. 2024b]	a set returns several options, not one decision; a risk-controlled threshold returns one
S4 training	a confidence the model is trained to state or to assign, per option for a typed decision [Yaldiz et al. 2026]	outcomes or rewards; training compute	none in general; one reward-model method adds conformal bounds [Pan et al. 2026a]	no: needs weights	a calibration and accuracy trade-off [Yaldiz et al. 2026]	can train per-option probabilities directly [Yaldiz et al. 2026]

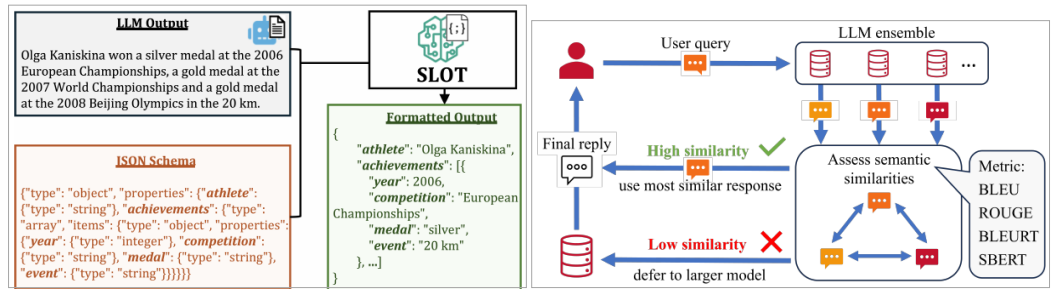
How confidence is produced and used, by stage (continued)



Gao et al. 2026

Fig. 4, p. 4

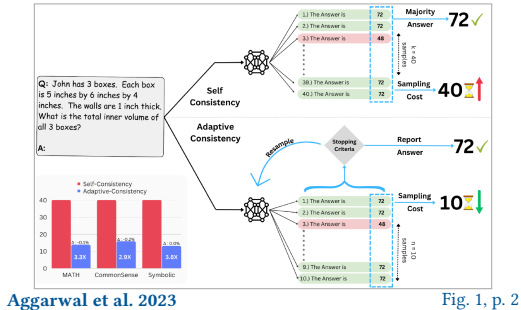
X consumers and contract



Wang et al. 2025e

Fig. 1, p. 1 Soiffer et al. 2025

Fig. 1, p. 1



Aggarwal et al. 2023

Fig. 1, p. 2

Fig. 8. How confidence is produced and used, by stage (continued).

2023b], or calibrates a set or a threshold with a conformal or risk-control guarantee [Campos et al. 2024]; 63 of the 66 works that state a guarantee sit here. The consumers (X, 71 works) route, defer, abstain, constrain decoding, or study how people rely on a stated confidence.

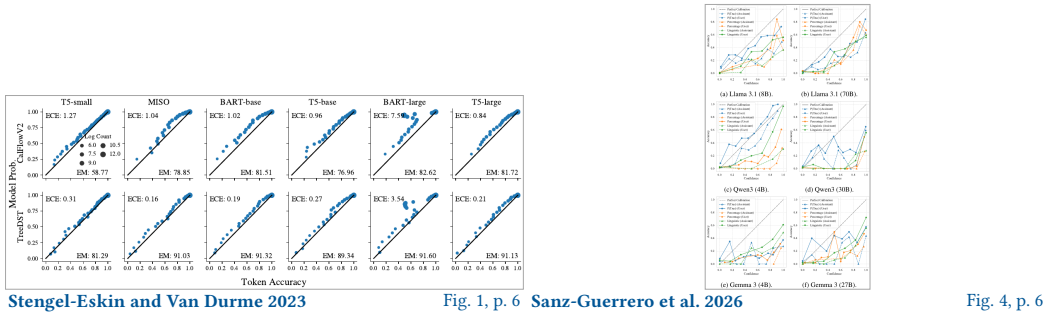
4 THE OUTPUT CONTRACT AND THE DECISION LOOP

A confidence is useful to software only if a rule can read it and act. Figure 10 draws the loop that this requires. A decision model receives a state and a *declared* set of options, assigns a probability to every option in one query, and passes the probabilities to an explicit action policy. The five stages of the taxonomy differ in where calibration enters this loop; the output contract determines how directly a rule can act on the result.

We record five contracts, the form of the output that a confidence is attached to (Figure 6b). With a *typed decision*, the options are declared before inference and the model assigns a probability to each, so that the decision and its confidence come from the same distribution and the action policy can

What calibration looks like in results, by stage

S0 read-off



S1 post-hoc map

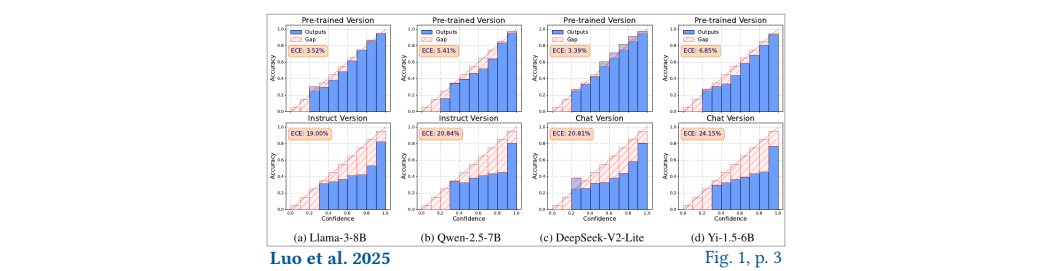
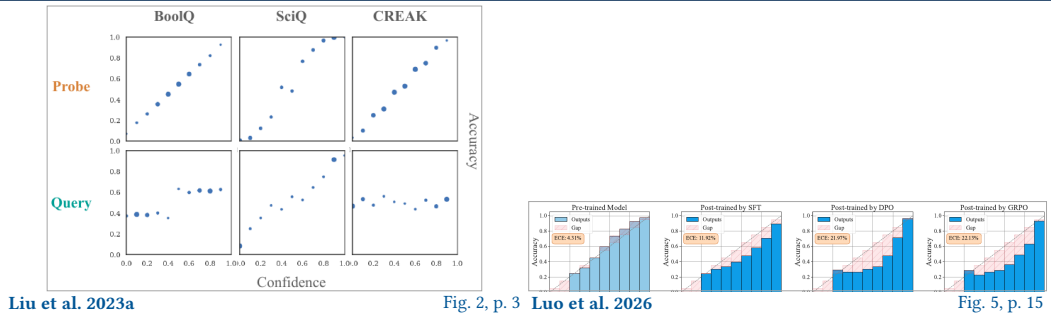
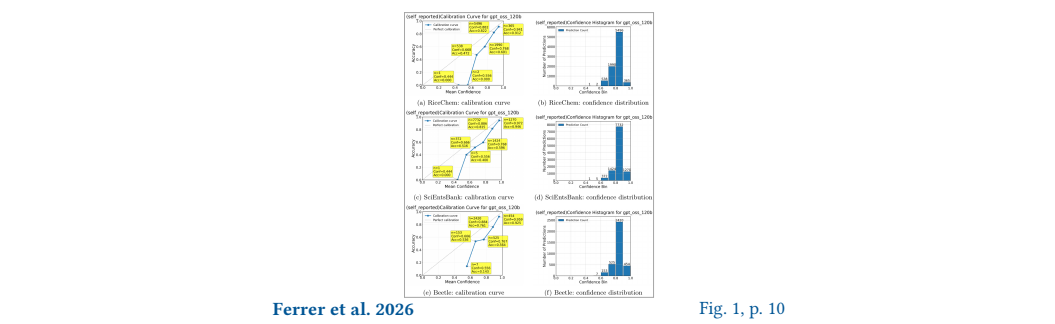
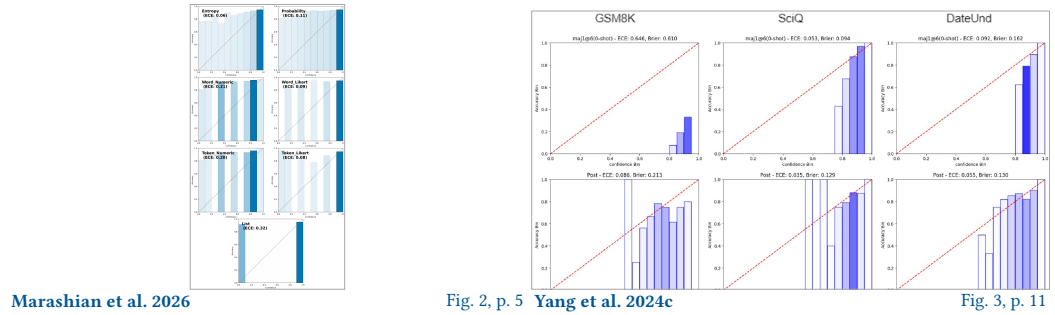


Fig. 9. **What calibration looks like in results, by stage.** Each panel shows a whole figure of the cited paper, reproduced under its CC BY licence; its label links to the paper. The selection rule and each panel’s figure, page and licence are in Table E1.

read the probability directly. A *set* returns several options with a coverage target. With *constrained output*, decoding is restricted to a grammar or schema; this fixes the form of the answer but not the judgement behind it, and it yields no confidence of its own. A *score* is one confidence number per answer, stated by the model, read off its token probabilities or estimated by a separate model. With *free text*, the answer is free text or a label with no per-answer probability, any confidence is a verbal hedge, an abstention, or the agreement of several sampled answers, and the answer must be parsed before a rule can use it. A probability that the model writes out as text, even for a declared outcome, is a score, not a typed decision. The contract of 1 work could not be determined from the paper.

Where the typed decision lives. 65 works attach the confidence to a typed decision, and 41 of them report a calibration metric (ECE, Brier score, NLL or a reliability diagram) in the main text. Most

What calibration looks like in results, by stage (continued)
S2 prompt elicitation



S3 extra inference

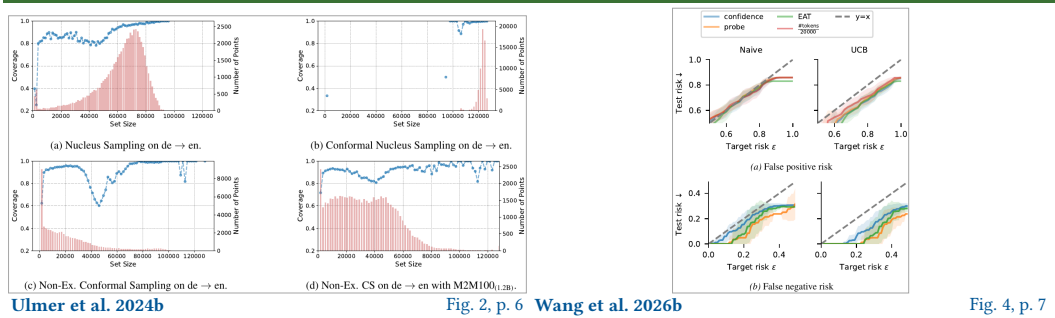
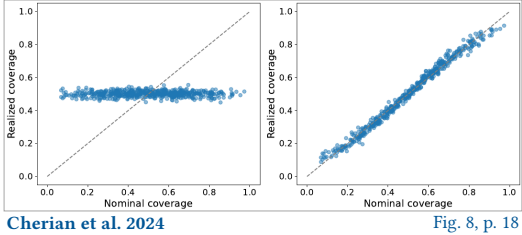


Fig. 9. What calibration looks like in results, by stage (continued).

sit at the earliest stages: 40 at the read-off and the post-hoc map, where the task is classification or multiple choice and the probability over the declared options is calibrated directly (24 of them report a calibration metric). A further 11 sit at extra inference and 7 among the consumers; none sit at prompt elicitation or among the theory works. The training core is where the typed decision is rarest: only 7 of the 80 core works train the probability of a declared option, all of which report its calibration. Nearly all the rest train a stated score (49 works) or a free-text answer with a hedge or a refusal (23). Table D1 (Appendix D) lists every typed work and the 28 consumers on grammar- or schema-constrained output; the constrained-output works mostly measure validity, speed, or accuracy under the constraint.

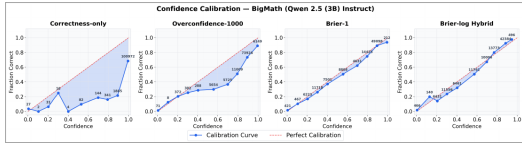
What calibration looks like in results, by stage (continued)



Cherian et al. 2024

Fig. 8, p. 18

S4 training



Tan et al. 2026

Fig. 4, p. 8 Steyvers et al. 2025a

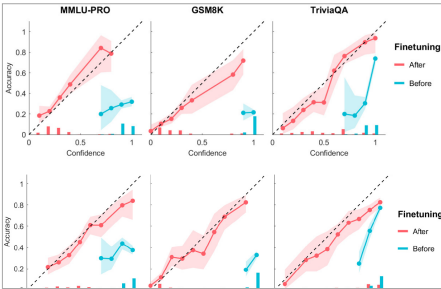
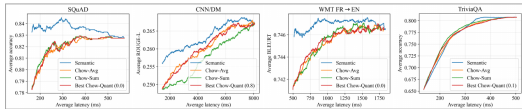


Fig. 3, p. 7

X consumers and contract



Soiffer et al. 2025

Fig. 2, p. 5 Fanconi and van der Schaar 2025

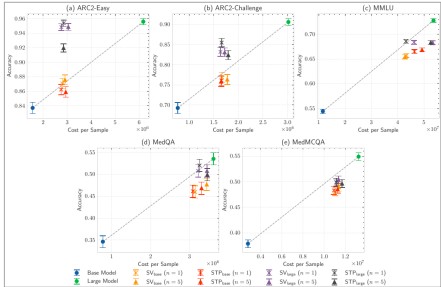
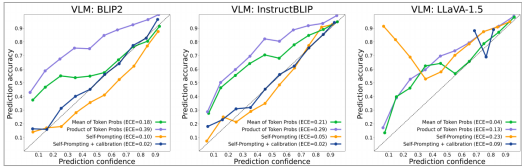


Fig. 3, p. 8



Srinivasan et al. 2024

Fig. 3, p. 5

Fig. 9. What calibration looks like in results, by stage (continued).

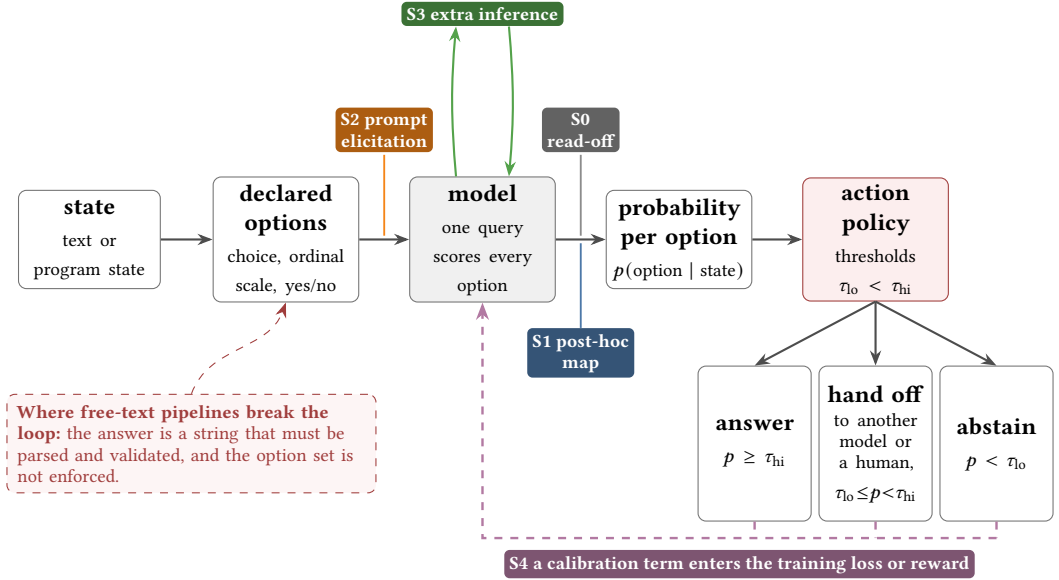


Fig. 10. **The calibrated decision loop, and where each stage injects calibration.** A decision model receives a state and a *declared* option set, scores every option in one query, and hands a probability per option to an explicit action policy. The policy drawn here answers when the probability is high, hands a less certain case off to another model or a human, and abstains when the probability is low; Table A1 lists the other low-confidence actions (trigger retrieval, filter the output, adapt compute, return a prediction set). The five stages of the taxonomy differ in where calibration enters: the model’s own read-off (S0), a post-hoc map (S1), a prompt that asks for confidence (S2), extra inference such as sampling or a conformal set (S3), or a calibration term in the model’s own training loss or reward (S4). The dashed callout marks where free-text generation pipelines break the loop (Table A1). Hosted and open typed-decision models (Tables 1 and 2) expose this interface: a state, declared options (a choice among labels, an ordinal scale or yes/no) and a probability per option.

What the confidence triggers. Across the corpus, most works stop at the confidence: for 259 of the 408 works no rule or trained behaviour acts on it. Where one does, the low-confidence action is to abstain (39 works), to hand the case off to another model or a person (35), to return a prediction set (30), to filter the output (20), to spend more computation (18), or to trigger retrieval (7). The threshold policy of Figure 10 therefore has to be supplied by whoever deploys the model, and the calibration of the probability it reads decides how well it works.

Hosted and open decision models. Some models are now offered with a typed-decision interface: unstructured state in, a probability per declared option out. The calibration of one such hosted model can be checked independently: a third-party evaluation is part of our corpus [Li et al. 2026c], and a public leaderboard measures the calibration of its chosen option. Tables 1 and 2 profile that model and the open decision models listed with it: the 268 repositories that a public model search for its name returns (Table 1), and the 68 systems of a community leaderboard [multimodalart 2026] that scores decision models on skill and on the calibration of the chosen option (Table 2).

5 A MEASURABLE AGENDA

Comparing the families of Section 3.1 is hard: their works are trained on different base models and data, report different metrics, and attach the confidence to different contracts. Table 4 proposes seven

Table 4. **A measurable agenda.** Seven quantities, each with a definition and a named public testbed, that together would let calibrated decision models be compared on common ground: calibration of the decision itself, safe abstention, cheap escalation, robustness to shift, the cost of a typed output contract, the size of a guaranteed set, and a controlled comparison of the training families. Figure 11 plots measured decision ECE (panels a and d) and the target shapes of selective risk and escalation regret (panels b and c).

Metric	Definition	Instantiation (testbed)	Stresses
Decision ECE	expected calibration error of the probability assigned to the chosen option within a <i>declared</i> option set, rather than of a free-text answer	MMLU as four-way choice [Hendrycks et al. 2021]; measured for every system on the Jev Decision Index (Table 2)	every stage; the main quantity for S4
Selective risk (AURC)	area under the risk-coverage curve obtained by abstaining below a confidence threshold [Geifman and El-Yaniv 2017]	LM-Polygraph benchmark [Vashurin et al. 2025]	S3, action policy
Escalation regret	extra cost, relative to the cheapest routing policy that reaches the same accuracy, when escalation to a larger model is triggered by confidence	RouterBench [Hu et al. 2024]	action policy, S1 to S4
Shift calibration gap	decision ECE on held-out subjects minus decision ECE on the subjects used for calibration or training	MMLU subject split, as in Shen et al. [2024]	S1, S4
Contract cost	schema-validity rate together with the accuracy change relative to unconstrained generation	JSONSchemaBench [Geng et al. 2025]; format-restriction study of Tam et al. [2024]	output contract
Guarantee tightness	average prediction-set size at a target coverage $1 - \alpha$	multiple-choice sets on MMLU [Hendrycks et al. 2021]	S3 (conformal)
Family comparison	decision ECE and accuracy of the five training families (T1 to T5) applied to the same base model and data	MMLU as four-way choice [Hendrycks et al. 2021], with the shared-base entries of the Jev Decision Index (Table 2) as a reference	S4 families

quantities, each with a definition and a named public testbed, that would make them comparable, and names the stage or family each one stresses.

Calibration of the decision itself. The first quantity, decision ECE, measures the calibration of the probability a model assigns to the option it chooses within a declared set, rather than of a free-text answer. It is the main quantity for the training stage, and it is the one quantity that this survey measures across many systems. Panels (a) and (d) of Figure 11 report it from a community leaderboard of decision models, the Jev Decision Index [multimodalart 2026] (release 0.2.1, a single snapshot of 2026-09-27, not peer reviewed), which scores 68 systems on skill over 38 decision benchmarks and on expected calibration error over a 1-in-6 sample of 32 of them. Skill and calibration separate on this board: the system ranked first on skill, Jev (57.91, against 57.44 for the next; a single point estimate), is not the best calibrated, and 17 of the 67 open systems have a lower ECE (Table 2). The board reports point estimates without uncertainty intervals and records no cost per decision, so it measures the first quantity only.

Acting on the confidence. Selective risk and escalation regret measure what the confidence buys once a rule acts on it: the error that remains after abstaining below a threshold [Geifman and El-Yaniv 2017], and the extra cost of escalating to a larger model when a cheaper policy would

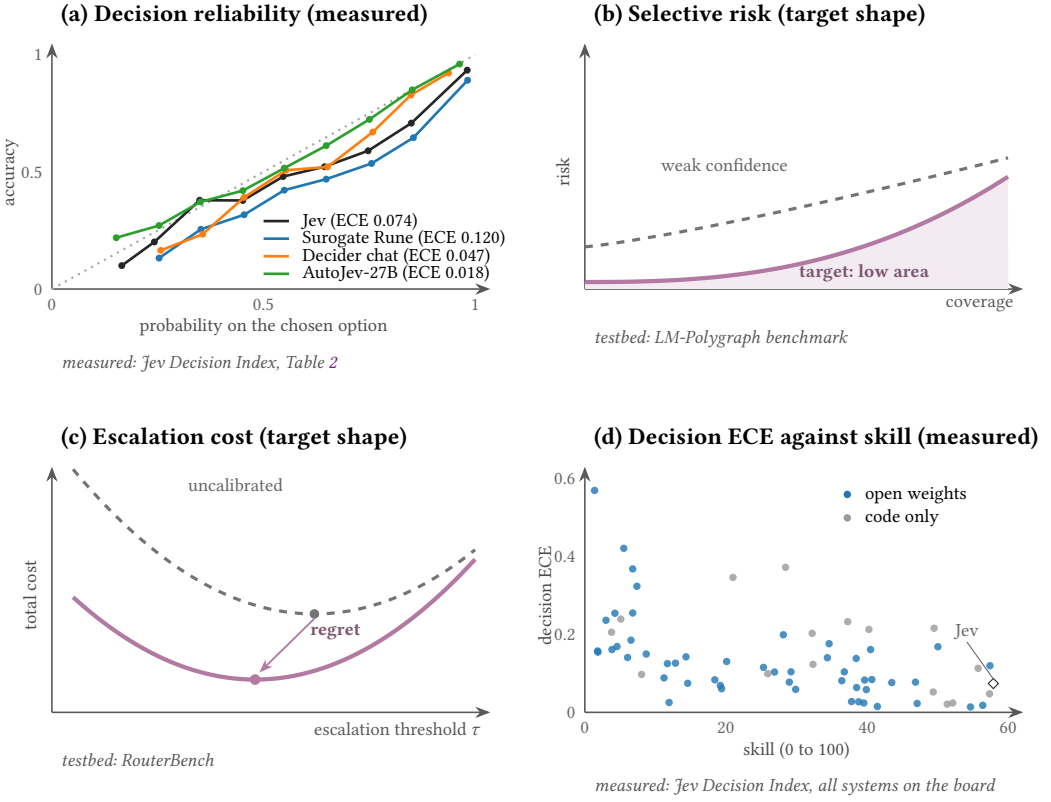


Fig. 11. **A measurable agenda, partly measured.** Decision ECE is the expected calibration error of the probability a model places on its chosen option (Table 4). Panels (a) and (d) come from one community leaderboard, the Jev Decision Index [multimodalart 2026] (release 0.2.1, a single snapshot of 2026-09-27; not peer reviewed): skill is its chance-normalised score over 38 decision benchmarks (0 to 100), ECE is computed on a 1-in-6 sample of 32 of them, and all values are point estimates without uncertainty intervals. (a) Reliability measured on the Jev Decision Index (Table 2) for Jev and the three open systems with the highest skill; each point is a confidence bin, and bins holding under 1% of answers are omitted. (b) Selective risk and (c) escalation regret are drawn as target shapes, not measured results: purple marks the target, grey dashed a typical baseline, and the named testbeds are in Table 4. (d) Decision ECE against skill for Jev and the 67 open systems on the board; lower right is better. The board records no cost per decision, so skill is used as the horizontal axis.

reach the same accuracy [Hu et al. 2024]. Panels (b) and (c) of Figure 11 draw their target shapes; they are not measured results.

Shift, contract and guarantee. The shift calibration gap asks whether a map or a trained confidence stays calibrated on subjects it was not fitted on [Shen et al. 2024]. The contract cost asks what a constrained or typed output costs in accuracy relative to free generation [Geng et al. 2025; Tam et al. 2024]. Guarantee tightness asks how large a prediction set must be to reach a target coverage. The last quantity, a comparison of the five training families on the same base model and data, is the controlled experiment that would say which training signal to prefer.

6 LIMITATIONS

Search. The query strings and the number of records screened were not recorded, so the search cannot be replayed exactly; Figure 4 marks those stages instead of estimating them. The 7 searches were run on one day, 2026-09-26, and the last year of the corpus is partial. Works placed only through the literature they cite, or not indexed by the searches, may be missing.

Placement. Stage, family and the per-work fields were assigned by reading each paper and were then checked against the papers in several review rounds and corrected where they disagreed. Some works sit on a boundary, for example a method that both trains a confidence and acts on it, and another reader could place them differently. The one-value-per-work fields (contract, action, metric) record the main contribution and the main text, not every variant a paper reports.

Evidence. The families are compared as the papers report them, on different base models, data and metrics; no claim here ranks one family above another. The leaderboard behind Figure 11 and Table 2 is a single snapshot, is not peer reviewed, and reports point estimates without uncertainty intervals. The gallery figures are limited to papers whose licence permits reuse, so they show only part of each stage.

7 CONCLUSION

Calibration of language models, long estimated after the fact, is increasingly also trained into the model. We organised 408 works by where calibration enters the decision pipeline, and the 80 training-stage works by what the training signal scores: supervised calibration objectives, reward-model calibration, calibration rewards on the stated confidence, abstention and refusal training, and process- and meaning-level confidence rewards. Crossing that typology with the output contract shows where the field is thin: the typed decision, whose probability is the very interface a program acts on, is calibrated after the fact in many works but trained in only 7 of the 80 training works; most train a stated score or a free-text hedge or refusal. The seven quantities of Table 4 would let the families be compared on common ground, and the first of them can already be measured.

COMPETING INTERESTS

The authors have no financial or other relationship with TypeSafe AI, the maker of Jev, or with the maintainer of the Jev Decision Index. Jev is profiled as one instance of a hosted typed-decision model.

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A OPERATIONAL DEFINITIONS

Table A1 gives the rule by which each work is placed: for every stage and property, what counts and the nearest case that does not.

Table A1. **Operational definitions used to place each work.** The stages (S0 to S4, plus the consumers X; theory and analysis works are listed but sit outside the core) are defined by *where the calibration signal enters the decision pipeline*, so a work is placed by what its main contribution changes, not by the task it is evaluated on. The first row defines a term, not an inclusion rule: works that report only discrimination are included and marked by their metric. The last three rows define properties recorded for every work: the Contract, Action and Guar. columns of Tables C1 and F1, where Action records what the rule or trained behaviour does when the confidence is low.

Construct	Counts if and only if...	Does not count
Calibrated confidence	the confidence is scored against empirical accuracy with a calibration metric (ECE, Brier, NLL or a reliability diagram)	a confidence that is only reported, or only used to rank answers (AUROC measures discrimination, not calibration)
S0 read-off	the confidence is the model's own token or sequence probability, unchanged	any map applied on top, including a probe on the hidden states (S1)
S1 post-hoc map	a map fitted on held-out data [Guo et al. 2017], on top of the read-off, as a probe on the hidden states of the frozen model, or as an auxiliary model reading the input and output text, or a fixed map (such as pooling the probability of synonymous labels or reweighting token probabilities), changes the confidence but not the model's parameters	re-training or fine-tuning the model (S4)
S2 prompt elicitation	the model is asked, in its input, to state a confidence in words or numbers [Tian et al. 2023]; a method whose main contribution is the prompt stays here even if it also aggregates several prompted answers	a confidence computed outside the model's own text (S0, S3)
S3 extra inference	the confidence needs extra inference, repeated sampling or perturbation (including internal-state signals compared across samples), or carries a distribution-free calibration: a prediction set or a threshold calibrated by conformal or risk control [Campos et al. 2024; Farquhar et al. 2024]	a single forward pass, read off (S0) or probed (S1)
S4 training	a calibration term appears in the model's own loss or reward: a confidence target or calibration loss, a calibrated reward model, a calibration reward, an abstention target, or a step- or meaning-level confidence reward (families T1 to T5) [Bani-Harouni et al. 2026b; Leng et al. 2025]	calibration that changes only as a side effect of training for something else; theory and analyses of such training, which are listed separately
X consumers and contract	the work uses a confidence to act (abstain, hand off, retrieve, filter, adapt compute, set), fixes the output contract, calibrates a model used as a judge against human labels, studies how people rely on a model's output or its stated confidence, or evaluates or benchmarks confidence-based abstention, without changing how the confidence is produced	a work whose main contribution is the confidence itself (S0 to S4)
Theory and analysis	the work analyses when or why training changes calibration, without proposing a training objective	a training method, even one motivated by such analysis (S4)

Table A1, continued

Construct	Counts if and only if...	Does not count
Typed decision output	the options are declared before inference and the model assigns a probability to each option, so the decision and its confidence come from the same distribution	a single output generated under a grammar or schema (constrained output); free text parsed after the fact
Action policy	an explicit rule, or a behaviour the model is trained into (such as refusing when unsure), maps the confidence to an action: answer when it is high; when it is low, abstain, hand the case off to another model or a human (which the papers call routing, cascading, deferral or escalation), trigger retrieval, filter the output (remove, rewrite or revise low-confidence claims, or keep the most confident of several sampled answers), adapt compute (keep sampling, reasoning, querying or exploring, or spend a larger budget; the same rule stops early when the confidence is high), or return a prediction set	a confidence reported with the answer or displayed to a person, with no rule or trained behaviour attached, or a selective-prediction curve reported only as an evaluation (recorded as none; the Metric column records the metrics of the main text)
Coverage guarantee	the method states a finite-sample, distribution-free bound, such as conformal coverage of at least $1 - \alpha$ [Campos et al. 2024]	good empirical calibration on a benchmark, without a stated bound

B PRIOR SURVEYS

Table B1 compares the 39 closest prior surveys on seven topics (Section 1.1).

Table B1. **Coverage of the 39 closest prior surveys:** every verified survey found by the search or by citation searching whose topic is calibration and uncertainty, reinforcement learning for language models, closed-set decisions, acting on confidence, hallucination or conformal prediction. CAL confidence estimation and calibration; RLC calibration-aware training of any kind; RLR reinforcement learning whose reward is a calibration or proper-scoring-rule objective; DEC decisions over a predefined option set; TYP typed, per-option decision outputs; ESC acting on confidence (abstention, deferral, routing); HAL hallucination. ✓ covered, ~ partial, – absent, ? not determinable. The RLR marks were checked in the full text or full reference list of every survey; the other marks come from abstracts and contents, corrected from a full-text reading for the surveys marked †. A 0 marks a column that no prior survey fully covers (TYP). Each full RLR mark is one subsection scoped to a single slice: 2024 methods [Li et al. 2025e], LLM agents [Lin et al. 2026] and forecasting [Xu et al. 2026b]. The last row marks the dimensions within this survey’s scope (●); hallucination is in scope only through abstention training and the theory works (○).

Survey	Venue	CAL	RLC	RLR	DEC	TYP	ESC	HAL
<i>Calibration and uncertainty quantification</i>								
Geng et al. [2024a]†	NAACL	✓	~	–	✓	–	~	~
Liu et al. [2025a]†	KDD	✓	~	–	–	–	~	~
Shorinwa et al. [2025]†	CSUR	✓	~	~	–	–	~	✓
Xia et al. [2025]	ACL Findings	✓	–	–	–	–	–	~
Kang et al. [2025a]†	arXiv	✓	–	–	–	–	~	✓
Zhang et al. [2026a]†	arXiv	✓	~	–	–	–	~	~
Wang [2023]†	arXiv	✓	–	–	✓	–	~	–
Li et al. [2025e]†	TMLR	✓	✓	✓	?	–	~	~
Lin et al. [2026]†	arXiv	✓	~	✓	?	~	✓	?
Li et al. [2025g]	ACL	✓	~	~	?	–	~	~
Xie et al. [2024b]	arXiv	✓	–	~	?	–	?	–
Abbasli et al. [2025]	IEEE TNNLS	✓	?	~	?	–	?	~
Huang et al. [2024c]	arXiv	✓	?	–	?	–	?	?
Beigi et al. [2024b]	arXiv	✓	?	–	?	–	?	?
Oh et al. [2026]	ACL	✓	~	–	?	–	~	?
Xia et al. [2026]	arXiv	✓	–	–	?	–	?	?
Liu et al. [2026b]	arXiv	✓	~	~	?	–	?	?
Sun et al. [2026b]	arXiv	✓	–	–	?	–	?	?
Xu et al. [2026b]†	arXiv	~	✓	✓	~	–	–	–
Ulmer et al. [2025]†	arXiv	✓	~	~	–	–	–	–
Zhang and Wang [2026]	J. Comput. Sci. Technol.	✓	?	–	~	–	?	?
He et al. [2026b]	Inf. Fusion	✓	?	–	–	–	?	?
<i>Reinforcement learning for LLMs</i>								
Zhang et al. [2025f]†	arXiv	–	–	~	–	–	–	–
Liu et al. [2025c]†	arXiv	–	–	–	–	–	–	–
Kaufmann et al. [2023]†	TMLR	–	?	–	–	–	–	–
Yu et al. [2026a]†	arXiv	–	?	–	–	–	–	–
<i>Closed-set decisions and judging</i>								
Gu et al. [2024]	arXiv	~	–	–	✓	–	–	–
Li et al. [2024c]	arXiv	~	–	~	✓	–	–	–
Li et al. [2024b]	arXiv	~	–	–	✓	–	–	–
<i>Acting on confidence (abstain, defer, route)</i>								
Wen et al. [2025]†	TACL	~	~	~	–	–	✓	~

Table B1, continued

Survey	Venue	CAL	RLC	RLR	DEC	TYP	ESC	HAL
Hendrickx et al. [2024]	Mach. Learn.	~	–	–	✓	–	✓	–
Strong et al. [2025]	Zenodo (preprint)	~	–	–	✓	–	✓	–
Moslem and Kelleher [2026][†]	TMLR	~	–	–	–	–	✓	–
Boudiaf et al. [2026][†]	arXiv	✓	–	~	~	–	✓	?
Hallucination								
Huang et al. [2025d]	ACM TOIS	~	–	–	–	–	–	✓
Alansari and Luqman [2026]	Comput. Sci. Rev.	~	–	~	–	–	–	✓
Zhang et al. [2025a][†]	arXiv	~	~	~	–	–	–	✓
Conformal prediction								
Campos et al. [2024][†]	TACL	✓	–	–	✓	–	~	–
Ashby et al. [2026]	Phil. Trans. A	✓	–	–	~	–	~	–
Surveys covering (✓)		24	2	3	8	0	6	5
This survey (scope)		●	●	●	●	●	●	○

C THE TRAINING CORE

Table C1 lists the 80 core works by training family, followed by the 5 theory and analysis works (Section 3.1).

Table C1. **The core: 80 works with a calibration term in the model’s own training loss or reward (S4)**, grouped by what the training signal scores; each work was assigned to one family by reading it. T1, the model is fitted to confidence targets or a calibration loss or regulariser, without a reward; T2, the reward model, a separate model in the training loop, is calibrated or made uncertainty-aware; T3, a reward or preference scores the stated confidence against the outcome, including proper scoring rules and ranking-based (AUC) confidence rewards, which target discrimination as well as calibration; T4, the signal targets the action: abstaining or refusing when unsure (supervised or rewarded); T5, the confidence reward is given per reasoning step or over meaning-equivalent answers. Family sizes: T1 23, T2 5, T3 31, T4 17, T5 4. The 5 theory and analysis works that follow are not counted in the core. * marks the 14 general-domain calibration-training works of 2025 and 2026 of which no prior survey cites more than one. Yr: year of the first public version (a citation may show a later venue year); Contract: the form of the output the confidence is attached to: typed (a probability per declared option), set (a prediction set), constrained (one output under a grammar or schema), score (one confidence number per answer, stated, read off or estimated by a separate model) or free-text (free text or a label with no per-answer probability; any confidence is a verbal hedge, an abstention, or agreement across sampled answers); Action: what an explicit rule or a trained behaviour does with the confidence when it is low (abstain; hand off = pass the case to another model or a human, which the papers call routing, cascading, deferral or escalation; retrieve = trigger retrieval; filter = remove, rewrite or revise low-confidence claims, or keep the most confident of several sampled answers; adapt compute = keep sampling, reasoning, querying or exploring, or spend a larger compute budget (the same rule stops early when the confidence is high); set = return a prediction set; none = no rule or trained behaviour acts on the confidence, including a selective-prediction curve reported only as an evaluation); Guar.: a stated distribution-free guarantee; Metric: the calibration or confidence metric reported in the main text (ECE, Brier, NLL, AUROC, selective = selective risk or accuracy at a coverage, coverage = achieved coverage of a prediction set or achieved risk under a controlled threshold, multiple = more than one of these (a reliability diagram counts with ECE; PRR with AUROC), other = a different metric, none = no confidence metric). ? = not determinable from the paper.

Work	Yr	Contribution	Contract	Action	Guar.	Metric
T1 supervised calibration objectives (23)						
Jiang et al. [2021]	2020	Finds T5, BART and GPT-2 QA probabilities are poorly calibrated and compares fine-tuning, temperature scaling and paraphrase fixes	typed	none	no	ECE
Kong et al. [2020]	2020	on- and off-manifold regularisers added to fine-tuning loss for in- and out-of-distribution calibration	typed	none	no	ECE
Lin et al. [2022]	2022	Fine-tunes GPT-3 to state calibrated numeric or word-level confidence without logits and introduces the CalibratedMath suite	score	none	no	multiple
Chaudhry et al. [2024]	2024	supervised fine-tuning on uncertainty-augmented answers to emit calibrated linguistic hedges	free-text	none	no	multiple
Chuang et al. [2024]	2024	Model emits a learned confidence token with each answer and low confidence triggers routing to a larger LLM	score	hand off	no	multiple
Han et al. [2024]	2024	LePe: fine-tunes on records of past answer correctness so the model learns to express its own confidence	score	none	no	ECE

Table C1, continued

Work	Yr	Contribution	Contract	Action	Guar.	Metric
Kapoor et al. [2024]	2024	calibration tuning on a small graded dataset of correct and incorrect answers yields generalising uncertainty estimates	score	none	no	multiple
Krishnan et al. [2024]	2024	uncertainty-aware causal LM loss grounded in decision theory raises uncertainty on wrong tokens	score	none	no	multiple
Liu et al. [2024c]	2024	self-training scheme in which the LLM learns to express uncertainty from its own consistency signals	score	retrieve	no	AUROC
Niu et al. [2024]	2024	UQ4CT: calibration loss over mixture-of-LoRA-experts functional space during PEFT	typed	none	no	ECE
Wang et al. [2024e]	2024	BLoB: jointly learns mean and covariance of LoRA weights by backpropagation for calibrated fine-tuning	typed	none	no	multiple
Zhang et al. [2024d]	2024	self-evaluation calibration tuning (SK-Tuning) then DPO on self-annotated factuality preferences	score	none	no	multiple
Eikema et al. [2025]	2025	Faithful Uncertainty Tuning: SFT on self-samples augmented with hedges matched to sample consistency	free-text	none	no	other
Hager et al. [2025]	2025	uncertainty distillation: SFT on verbalised probabilities derived from Monte Carlo semantic sampling	score	none	no	multiple
Huang et al. [2025a]	2025	Self-Calibration: distills self-consistency confidence into a single-pass confidence for efficient test-time scaling	score	adapt compute	no	multiple
Jang et al. [2025]	2025	SFT on scalar confidence labels alone elicits self-verification in chain-of-thought	score	adapt compute	no	multiple
Li et al. [2025d]*	2025	ConfTuner: tokenised Brier score loss (a proper scoring rule) for verbalised confidence fine-tuning	score	hand off	no	multiple
Steyvers et al. [2025a]	2025	tests cross-domain transfer of supervised fine-tuning for numeric confidence and pairwise confidence comparison	score	none	no	multiple
Xiao et al. [2025]	2025	calibration-aware fine-tuning plus EM-based ECE regulariser to restore calibration lost in preference alignment	typed	none	no	ECE
Zhang et al. [2025c]	2025	Direct Confidence Alignment: DPO aligns verbalised confidence with internal token-probability confidence	score	none	no	other
Zhang et al. [2025b]	2025	GrACE: fine-tunes a special confidence token whose hidden-state similarity gives calibrated confidence	score	adapt compute	no	multiple
Guo et al. [2026]	2026	post-trains end-of-reasoning verbalised confidence and an in-reasoning <uncertain> marker used to trigger RAG	score	retrieve	no	multiple
Xie et al. [2026]	2026	post-RL SFT with self-distillation restores anchor-token confidence reliability degraded by GRPO and DPO	typed	retrieve	no	multiple
T2 reward-model calibration (5)						
Leng et al. [2025]	2024	PPO-M calibrated reward modelling with confidence-augmented pairs and PPO-C reward adjustment by running mean	score	none	no	multiple

Table C1, continued

Work	Yr	Contribution	Contract	Action	Guar.	Metric
Lou et al. [2024]	2024	URM: reward model outputs mean and variance of attribute rewards to expose reward uncertainty	score	none	no	selective
Park et al. [2025a]	2025	quantile-regression calibration of process reward models used for instance-adaptive compute	score	adapt compute	no	multiple
Ma et al. [2026a]	2026	conditional optimal transport over PRM hidden states yields calibrated success quantiles	score	adapt compute	no	multiple
Pan et al. [2026a]	2026	UARM: quantile conformal uncertainty for reward models plus variance-reweighted GRPO advantages	set	none	yes	selective
T3 calibration rewards on the stated confidence (31)						
Band et al. [2024]	2024	linguistic calibration: SFT then RL rewarding generations that let a simulated reader make calibrated forecasts	free-text	none	no	ECE
Stengel-Eskin et al. [2024]	2024	listener-aware DPO where preferences reward answers a simulated listener accepts only when correct	free-text	abstain	no	multiple
Tao et al. [2024]	2024	CONQORD: PPO with quality reward plus order-preserving reward aligning verbalised confidence with response quality	score	retrieve	no	ECE
Xu et al. [2024a]	2024	SaySelf: SFT on self-reflective rationales then RL with a reward penalising overconfident wrong answers	score	none	no	ECE
Bani-Harouni et al. [2026b]*	2025	Rewarding Doubt: log scoring rule reward on verbalised confidence	score	none	no	multiple
Damani et al. [2025]*	2025	RLCR: binary correctness reward plus Brier score reward on reasoned confidence	score	none	no	multiple
Kale and Dhami [2025]	2025	KnowRL: introspection plus consensus-based self-reward to strengthen feasibility self-knowledge	free-text	none	no	other
Liu et al. [2025b]	2025	C2GSPG: group sequence policy gradient with a cross-entropy regulariser aligning sequence confidence with reward	score	none	no	multiple
Turtel et al. [2025]	2025	outcome-based RLVR on resolved prediction-market events that improves forecast calibration	score	none	no	multiple
van Niekerk et al. [2025]	2025	RLSF: preference optimisation using the frozen model own answer-span confidence as self-feedback reward	score	none	no	ECE
Wu et al. [2025a]*	2025	behaviourally calibrated RL with strictly proper scoring rule rewards enabling abstention or claim flagging	score	abstain	no	multiple
Yin et al. [2026]	2025	Deliberative Searcher: search agent RL maximising accuracy under a soft reliability (confidence) constraint	score	none	no	other
Zhang et al. [2025e]	2025	LoVeC: GRPO and DPO with calibration reward for sentence-level confidence tags in long-form output	score	none	no	multiple
Zou et al. [2025]	2025	GRPO with composite reward penalising unjustified high confidence and token-entropy spikes	score	none	no	multiple

Table C1, continued

Work	Yr	Contribution	Contract	Action	Guar.	Metric
Bani-Harouni et al. [2026a]	2026	ConRad: RL with calibration reward for report- and sentence-level verbalised confidence in radiology LVLMS	score	none	no	multiple
Cui et al. [2026]	2026	RAC: ranking-aware group loss and clean-corrupted pairwise confidence loss inside multimodal RL	score	none	no	multiple
Li et al. [2026d]	2026	CoCA: confidence-first GRPO with segmented Brier-style confidence reward and answer reward	score	none	no	multiple
Li et al. [2026a]	2026	ORCE: rank-based RL reward aligning post-hoc verbalised confidence order with sampled correctness surrogate	score	none	no	multiple
Liu et al. [2026a]	2026	RLMF: preference optimisation whose completion rankings use metacognitive feedback on self-judgement quality	score	none	no	Brier
Liu and Wang [2026]	2026	CURE: claim-level confidence reasoning trained with DPO confidence correction and RL	score	filter	no	multiple
Ma et al. [2026b]*	2026	DCPO: decouples reasoning and calibration objectives with separate advantages to avoid gradient conflict	score	none	no	multiple
Singh et al. [2026b]	2026	studies verifiable Brier rewards for aleatoric forecasting (NFL win probability) and why per-outcome reward fails	score	none	no	multiple
Tan et al. [2026]*	2026	characterises non-hackable confidence reward schemes (Brier, log hybrids) against confidence reward hacking	score	none	no	multiple
Turtel et al. [2026]	2026	compares five proper scoring rules (Brier, log, others) as terminal RL rewards for LLM forecasters	score	none	no	multiple
Wang et al. [2026a]*	2026	CAPO: logistic AUC surrogate loss for uncertainty-aware advantage estimation in GRPO	score	none	no	multiple
Xiao et al. [2026]	2026	VL-Calibration: RL rewarding separate visual and reasoning confidence in LVLMS	score	none	no	multiple
Xu et al. [2026a]	2026	DualStake: margin-clipped confidence-dependent stake rewards for evidence and answer confidence in deep research agents	score	none	no	multiple
Xuan et al. [2026]	2026	RL fine-tuning jointly rewarding task accuracy and verbalised calibration in tool-use agents	score	none	no	multiple
Yaldiz et al. [2026]*	2026	calibration-aware RLVR that directly adjusts decision-token probabilities to curb overconfidence	typed	none	no	ECE
Yang et al. [2026b]	2026	C3RL: correctness reward plus calibration reward plus dataset reference-accuracy reward for adaptive test-time scaling	score	adapt compute	no	multiple
Zhu [2026]	2026	RefGRPO: free calibration bonus contrasting agent reflection score with actual environment outcome	score	filter	no	selective

Table C1, continued

Work	Yr	Contribution	Contract	Action	Guar.	Metric
T4 abstention and refusal training (17)						
Yang et al. [2024b]	2023	alignment for honesty: fine-tuning to answer known questions and say I do not know otherwise, with honesty metrics	free-text	abstain	no	other
Zhang et al. [2023a]	2023	R-Tuning: builds refusal-aware SFT data by appending uncertainty expressions to questions the model gets wrong	score	abstain	no	other
Chen et al. [2025a]	2024	CoKE: probes knowledge boundary via internal confidence then trains the model to express it	free-text	abstain	no	other
Cheng et al. [2024]	2024	Idk dataset built per model and SFT plus preference optimisation to refuse unknown questions	free-text	abstain	no	other
Kang et al. [2025b]	2024	shows unfamiliar finetuning examples shape hallucination and proposes more reliable reward models for RL factuality finetuning	free-text	abstain	no	other
Li et al. [2025c]	2024	US-Tuning: two-stage instruction tuning for knowledge-boundary awareness then instruction adherence	free-text	abstain	no	other
Liang et al. [2024a]	2024	RLKF: knowledge-state preference data from DreamCatcher used as reward for honesty-oriented RL	free-text	abstain	no	other
Xu et al. [2024b]	2024	RLKF with a reliability reward model trained on knowledge-feedback preferences to refuse out-of-knowledge questions	free-text	abstain	no	other
Xue et al. [2025]	2024	UAlign: uncertainty estimates (confidence, semantic entropy) as features for SFT and PPO factuality alignment	free-text	abstain	no	AUROC
Mohamadi et al. [2025]*	2025	Reinforced Hesitation: ternary RLVR reward (+1, 0, -lambda) with penalty-dependent abstention frontier	free-text	abstain	no	other
Song et al. [2025]	2025	shows reinforcement finetuning erodes refusal and restores it by mixing synthetic unanswerable math	free-text	abstain	no	other
Wei et al. [2025]*	2025	TruthRL: ternary reward (correct +1, abstain 0, hallucination -1) under GRPO	free-text	abstain	no	other
Che et al. [2026]	2026	proves error-penalised abstention rewards can collapse KL-anchored RL to refuse-all and proposes a repair	score	abstain	no	Brier
Gao et al. [2026]*	2026	KARL: knowledge-boundary-aware reward from within-group statistics plus two-stage RL	free-text	abstain	no	other
Pan et al. [2026b]	2026	TIAR: trajectory-informed reweighting of the abstention advantage within GRPO ternary reward	free-text	abstain	no	other
Song et al. [2026]	2026	Visual-Idk dataset and SFT plus preference optimisation so VLMs refuse unknown visual facts	free-text	abstain	no	other
Zhai et al. [2026]*	2026	Abstain-R1: clarification-aware RLVR reward for explicit abstention plus post-refusal clarification	free-text	abstain	no	other

Table C1, continued

Work	Yr	Contribution	Contract	Action	Guar.	Metric
T5 process- and meaning-level confidence rewards (4)						
An and Xu [2025]	2025	FiSCoRe: fine-grained sample-specific semantic confidence reward guiding abstention in GRPO	free-text	abstain	no	other
He et al. [2025]	2025	MMBoundary: SFT on step-level self-rewarded confidence then RL with step calibration rewards for MLLMs	free-text	none	no	ECE
Wang et al. [2026d]*	2026	RLCM: margin-enhanced process reward on confidence at intermediate reasoning budgets plus Brier term	score	abstain	no	multiple
Yu et al. [2026b]*	2026	CSR: correctness reward plus semantic calibration reward on agreement among rollouts, no verbal confidence	free-text	none	no	multiple
Theory and analysis of training-stage calibration (not counted in the core) (5)						
Blasiok et al. [2023]	2023	relates proper-loss local optimality to smooth calibration, explaining when training yields calibration	?	none	no	none
Kalai and Vempala [2024]	2023	proves calibrated LMs must hallucinate arbitrary facts at a rate near the Good-Turing fraction of facts seen once	free-text	none	no	none
Bereket and Leskovec [2025]	2025	shows GRPO group-std normalisation causes overconfidence on stochastic outcomes; PPO and RLOO stay calibrated	score	none	no	multiple
Kalai et al. [2025]	2025	argues binary-graded training and evaluation reward guessing and calls for scoring that credits admitting uncertainty	free-text	none	no	none
Nakkiran et al. [2025]	2025	theory that semantic calibration emerges from next-token loss and breaks under RL tuning and CoT	free-text	none	no	ECE

D THE OUTPUT CONTRACT

Table D1 lists every typed work and the consumers on grammar- or schema-constrained output (Section 4).

Table D1. **The output contract: 93 works**, selected by a stated rule: every work whose confidence is attached to a typed decision (contract “typed”, 65, at any stage), and the X-stage works on grammar- or schema-constrained output (28). Cal.: ✓ when the main text reports a calibration metric (ECE, Brier score, NLL or a reliability diagram); 41 of the 65 typed works do, all 7 in the training core. The constrained-output works mostly measure validity, speed or accuracy under the constraint. Each work’s contribution is in Table C1 or Table F1. Yr: year of the first public version; Family: the training family (S4) or method family; Metric: the calibration or confidence metric reported in the main text (ECE, Brier, NLL, AUROC, selective = selective risk or accuracy at a coverage, coverage = achieved coverage of a prediction set or achieved risk under a controlled threshold, multiple = more than one of these (a reliability diagram counts with ECE; PRR with AUROC), other = a different metric, none = no confidence metric). ? = not determinable from the paper.

Work	Yr	Family	Metric	Cal.
Typed output, S0 read-off (16)				
OpenAI [2023]	2023	Calibration after alignment	ECE	✓
Huang et al. [2026]	2026	Calibration after alignment	ECE	✓
Proskurina et al. [2026]	2026	Calibration after alignment	ECE	✓
Sanz-Guerrero et al. [2026]	2026	Calibration after alignment	multiple	✓
Desai and Durrett [2020]	2020	Empirical studies of read-off calibration	ECE	✓
Dan and Roth [2021]	2021	Empirical studies of read-off calibration	ECE	✓
Ahuja et al. [2022]	2022	Empirical studies of read-off calibration	ECE	✓
Chen et al. [2023a]	2022	Empirical studies of read-off calibration	ECE	✓
Xiao et al. [2022]	2022	Empirical studies of read-off calibration	multiple	✓
Li et al. [2025h]	2023	Empirical studies of read-off calibration	multiple	✓
Zhang et al. [2023d]	2023	Empirical studies of read-off calibration	ECE	✓
Mahaut et al. [2024]	2024	Empirical studies of read-off calibration	other	–
Joshi et al. [2025]	2025	Empirical studies of read-off calibration	ECE	✓
Lovering et al. [2025]	2024	Likelihood and logit read-off	other	–
Plaut et al. [2024]	2024	Likelihood and logit read-off	multiple	✓
Wang et al. [2024d]	2024	Likelihood and logit read-off	none	–
Typed output, S1 post-hoc map (24)				
Badhe et al. [2026]	2026	Fixed maps over the read-off	multiple	✓
Zheng et al. [2023]	2023	Label-bias and option-prior debiasing	none	–
Reif and Schwartz [2024]	2024	Label-bias and option-prior debiasing	none	–
Han et al. [2022]	2022	Learned calibrators	none	–
Luo et al. [2026]	2026	Learned calibrators	multiple	✓
Liu et al. [2024b]	2024	Multicalibration and task-specific recalibration	ECE	✓
Burns et al. [2022]	2022	Probes and learned confidence heads	other	–
Azaria and Mitchell [2023]	2023	Probes and learned confidence heads	AUROC	–
Liu et al. [2023a]	2023	Probes and learned confidence heads	ECE	✓
Holtzman et al. [2021]	2021	Prompt and context calibration	none	–
Zhao et al. [2021]	2021	Prompt and context calibration	none	–
Fei et al. [2023]	2023	Prompt and context calibration	none	–
Jiang et al. [2023b]	2023	Prompt and context calibration	none	–
Liusie et al. [2023]	2023	Prompt and context calibration	none	–
Zhou et al. [2023b]	2023	Prompt and context calibration	none	–
Abbas et al. [2024]	2024	Prompt and context calibration	ECE	✓
Cho et al. [2025]	2024	Prompt and context calibration	none	–
Gundem et al. [2025]	2025	Prompt and context calibration	none	–
Guo et al. [2017]	2017	Scaling and binning	multiple	✓

Table D1, continued

Work	Yr	Family	Metric	Cal.
Kull et al. [2019]	2019	Scaling and binning	multiple	✓
Kumar et al. [2019]	2019	Scaling and binning	multiple	✓
Gupta and Ramdas [2021]	2021	Scaling and binning	ECE	✓
Shen et al. [2024]	2024	Scaling and binning	multiple	✓
Luo et al. [2025]	2025	Scaling and binning	multiple	✓
Typed output, S3 extra inference (11)				
Ling et al. [2024]	2024	Benchmarks and empirical studies	AUROC	–
Jung et al. [2024]	2024	Conformal abstention, selection and cascades	multiple	✓
Tonolini et al. [2024]	2024	Ensembles and cross-model disagreement	multiple	✓
Kruse et al. [2025]	2025	Ensembles and cross-model disagreement	multiple	✓
Schuster et al. [2021]	2021	Risk control and early exit	coverage	–
Jazbec et al. [2024]	2024	Risk control and early exit	multiple	✓
Liu et al. [2023b]	2023	Sampling consistency	other	–
Portillo Wightman et al. [2023]	2023	Sampling consistency	ECE	✓
Becker and Soatto [2024]	2024	Sampling consistency	multiple	✓
Yang et al. [2024a]	2024	Sampling consistency	multiple	✓
Zhang et al. [2024a]	2024	Sampling consistency	ECE	✓
Typed output, S4 training (7)				
Yaldiz et al. [2026]	2026	T3 calibration rewards on the stated confidence	ECE	✓
Kong et al. [2020]	2020	T1 supervised calibration objectives	ECE	✓
Niu et al. [2024]	2024	T1 supervised calibration objectives	ECE	✓
Wang et al. [2024e]	2024	T1 supervised calibration objectives	multiple	✓
Xiao et al. [2025]	2025	T1 supervised calibration objectives	ECE	✓
Xie et al. [2026]	2026	T1 supervised calibration objectives	multiple	✓
Jiang et al. [2021]	2020	T1 supervised calibration objectives	ECE	✓
Typed output, X consumers and contract (7)				
Jitkrittum et al. [2023]	2023	Deferral	none	–
Kondadadi and Ortega [2026]	2026	Deferral	none	–
Li et al. [2026c]	2026	Judge calibration	multiple	✓
Varshney and Baral [2022]	2022	Routing/cascade	none	–
Ramirez et al. [2024]	2024	Routing/cascade	none	–
Zellinger and Thomson [2025]	2025	Routing/cascade	ECE	✓
Wang et al. [2024c]	2024	Typed classification	none	–
Constrained output, X consumers and contract (28)				
Scholak et al. [2021]	2021	Constrained decoding	none	
Beurer-Kellner et al. [2023]	2022	Constrained decoding	none	
Poesia et al. [2022]	2022	Constrained decoding	none	
Geng et al. [2023]	2023	Constrained decoding	none	
Lew et al. [2023]	2023	Constrained decoding	none	
Willard and Louf [2023]	2023	Constrained decoding	none	
Beurer-Kellner et al. [2024]	2024	Constrained decoding	none	
Dong et al. [2025]	2024	Constrained decoding	none	
Geng et al. [2024b]	2024	Constrained decoding	none	
Koo et al. [2024]	2024	Constrained decoding	none	
Park et al. [2024]	2024	Constrained decoding	none	
Ugare et al. [2024]	2024	Constrained decoding	none	
Banerjee et al. [2025]	2025	Constrained decoding	none	
Li et al. [2026b]	2025	Constrained decoding	none	
Loula et al. [2025]	2025	Constrained decoding	other	
Mündler et al. [2025]	2025	Constrained decoding	none	
Park et al. [2025b]	2025	Constrained decoding	none	
Reddy et al. [2026]	2026	Constrained decoding	none	

Table D1, continued

Work	Yr	Family	Metric	Cal.
Tang et al. [2023]	2023	Structured-output benchmark	none	
Shorten et al. [2024]	2024	Structured-output benchmark	none	
Tam et al. [2024]	2024	Structured-output benchmark	none	
Xia et al. [2024a]	2024	Structured-output benchmark	none	
Geng et al. [2025]	2025	Structured-output benchmark	none	
Yang et al. [2025a]	2025	Structured-output benchmark	none	
Fan [2026]	2026	Structured-output benchmark	none	
Galeone et al. [2026]	2026	Structured-output benchmark	none	
Lee et al. [2026a]	2026	Structured-output benchmark	none	
Singh et al. [2026a]	2026	Structured-output benchmark	none	

E PROVENANCE OF THE GALLERY PANELS

Table E1 gives, for every panel of Figures 1, 8 and 9, the source figure and page in the cited paper and the licence under which it is reproduced, together with the rule by which panels were chosen.

Table E1. **Provenance of the gallery panels** in Figures 1, 8 and 9: the source figure and page in the cited arXiv paper, and the licence under which it is reproduced. Panels are chosen by a fixed rule: only papers whose licence permits reuse (CC BY); per stage, one panel per method family (per training family at S4), taking the largest families first and, in each, the newest figure that legibly shows the method or a result, at most 3 per stage; families with no reusable figure that legibly shows the method or a result are skipped. Figure 8: No reusable figure passed the visual check for S0 read-off. At S4, T5 had a figure but was cut by the cap of 3. T2 has no reusable figure that passed the check. Figure 9: At S4, T2, T4, T5 have no reusable figure that passed the check.

Work	Source	Panel	Licence	Used in Fig.
S0 read-off				
Sanz-Guerrero et al. [2026]	arXiv:2606.03437	Fig. 4, p. 6	CC BY 4.0	9
Stengel-Eskin and Van Durme [2023]	arXiv:2211.07443	Fig. 1, p. 6	CC BY 4.0	1, 9
S1 post-hoc map				
Gundem et al. [2025]	arXiv:2505.23783	Fig. 2, p. 5	CC BY 4.0	8
Liu et al. [2023a]	arXiv:2312.03729	Fig. 2, p. 3	CC BY 4.0	9
Lu [2026]	arXiv:2607.16799	Fig. 1, p. 2	CC BY 4.0	1, 8
Luo et al. [2025]	arXiv:2505.16690	Fig. 1, p. 3	CC BY 4.0	9
Luo et al. [2026]	arXiv:2601.04277	Fig. 1, p. 2	CC BY 4.0	8
Luo et al. [2026]	arXiv:2601.04277	Fig. 5, p. 15	CC BY 4.0	9
S2 prompt elicitation				
Ferrer et al. [2026]	arXiv:2603.29559	Fig. 1, p. 10	CC BY 4.0	9
Kumar et al. [2024]	arXiv:2405.16282	Fig. 1, p. 1	CC BY 4.0	8
Marashian et al. [2026]	arXiv:2606.17234	Fig. 1, p. 1	CC BY 4.0	1, 8
Marashian et al. [2026]	arXiv:2606.17234	Fig. 2, p. 5	CC BY 4.0	9
Yang et al. [2024c]	arXiv:2404.09127	Fig. 3, p. 11	CC BY 4.0	9
S3 extra inference				
Chen et al. [2025b]	arXiv:2502.20285	Fig. 3, p. 8	CC BY 4.0	8
Cherian et al. [2024]	arXiv:2406.09714	Fig. 8, p. 18	CC BY 4.0	9
Cohen et al. [2023]	arXiv:2305.13281	Fig. 2, p. 3	CC BY 4.0	8
Doula et al. [2025]	arXiv:2505.09427	Fig. 1, p. 3	CC BY 4.0	1, 8
Ulmer et al. [2024b]	arXiv:2402.00707	Fig. 2, p. 6	CC BY 4.0	9
Wang et al. [2026b]	arXiv:2602.03814	Fig. 4, p. 7	CC BY 4.0	9
S4 training				
Gao et al. [2026]	arXiv:2604.22779	Fig. 4, p. 4	CC BY 4.0	1, 8
Guo et al. [2026]	arXiv:2604.05306	Fig. 1, p. 2	CC BY 4.0	1, 8
Steyvers et al. [2025a]	arXiv:2510.05126	Fig. 3, p. 7	CC BY 4.0	9
Tan et al. [2026]	arXiv:2607.04332	Fig. 4, p. 8	CC BY 4.0	9
Yang et al. [2026b]	arXiv:2607.01612	Fig. 1, p. 3	CC BY 4.0	1, 8
X consumers and contract				
Aggarwal et al. [2023]	arXiv:2305.11860	Fig. 1, p. 2	CC BY 4.0	8
Fanconi and van der Schaar [2025]	arXiv:2506.11887	Fig. 3, p. 8	CC BY 4.0	9
Soiffer et al. [2025]	arXiv:2509.21837	Fig. 1, p. 1	CC BY 4.0	8
Soiffer et al. [2025]	arXiv:2509.21837	Fig. 2, p. 5	CC BY 4.0	9
Srinivasan et al. [2024]	arXiv:2402.15610	Fig. 3, p. 5	CC BY 4.0	9
Wang et al. [2025e]	arXiv:2505.04016	Fig. 1, p. 1	CC BY 4.0	8

F THE LANDSCAPE

Table F1 lists the 323 works outside the training core, grouped by the stage at which calibration enters and then by method family. The 80 core works and the 5 theory and analysis works are in Table C1.

Table F1. **The landscape: the 323 works outside the training core**, grouped by the stage at which calibration enters (Table A1) and then by method family, oldest first; the 80 core works and the 5 theory and analysis works are listed in Table C1. Yr: year of the first public version (a citation may show a later venue year); Contract: the form of the output the confidence is attached to: typed (a probability per declared option), set (a prediction set), constrained (one output under a grammar or schema), score (one confidence number per answer, stated, read off or estimated by a separate model) or free-text (free text or a label with no per-answer probability; any confidence is a verbal hedge, an abstention, or agreement across sampled answers); Access: white (needs logits or weights), black (text only) or either; Samples: the method needs repeated inference; Guar.: a stated distribution-free guarantee; Trained: yes when calibration is a training objective of the model itself, partial when a separate model (a probe, calibrator, router or reward model) is trained; Metric: the calibration or confidence metric reported in the main text (ECE, Brier, NLL, AUROC, selective = selective risk or accuracy at a coverage, coverage = achieved coverage of a prediction set or achieved risk under a controlled threshold, multiple = more than one of these (a reliability diagram counts with ECE; PRR with AUROC), other = a different metric, none = no confidence metric); Task: the task evaluated; Action: what an explicit rule or a trained behaviour does with the confidence when it is low (abstain; hand off = pass the case to another model or a human, which the papers call routing, cascading, deferral or escalation; retrieve = trigger retrieval; filter = remove, rewrite or revise low-confidence claims, or keep the most confident of several sampled answers; adapt compute = keep sampling, reasoning, querying or exploring, or spend a larger compute budget (the same rule stops early when the confidence is high); set = return a prediction set; none = no rule or trained behaviour acts on the confidence, including a selective-prediction curve reported only as an evaluation). ? = not determinable from the paper.

Work	Yr	Contract	Access	Samples	Guar.	Trained	Metric	Task	Action
S0 read-off (25)									
<i>Calibration after alignment</i>									
OpenAI [2023]	2023	typed	white	no	no	no	ECE	QA	none
Zhu et al. [2023]	2023	score	white	no	no	no	ECE	QA	none
Huang et al. [2026]	2026	typed	white	no	no	no	ECE	QA	none
Proskurina et al. [2026]	2026	typed	white	no	no	no	ECE	QA	none
Sanz-Guerrero et al. [2026]	2026	typed	white	no	no	no	multiple	QA	none
<i>Empirical studies of read-off calibration</i>									
Desai and Durrett [2020]	2020	typed	white	no	no	no	ECE	classification	none
Wang et al. [2020]	2020	score	white	no	no	no	ECE	generation	none
Dan and Roth [2021]	2021	typed	white	no	no	no	ECE	classification	none
Xiao and Wang [2021]	2021	score	white	no	no	no	other	generation	none
Ahuja et al. [2022]	2022	typed	white	no	no	no	ECE	classification	none
Chen et al. [2023a]	2022	typed	white	no	no	no	ECE	classification	none
Si et al. [2022]	2022	score	white	no	no	no	ECE	QA	none
Stengel-Eskin and Van Durme [2023]	2022	constrained	white	no	no	no	ECE	code	none
Xiao et al. [2022]	2022	typed	white	no	no	no	multiple	classification	none
Li et al. [2025h]	2023	typed	white	no	no	no	multiple	classification	none
Zhang et al. [2023d]	2023	typed	white	no	no	no	ECE	classification	none
Mahaut et al. [2024]	2024	typed	either	no	no	no	other	QA	none
Stolfo et al. [2024]	2024	score	white	no	no	no	other	generation	none
Joshi et al. [2025]	2025	typed	white	no	no	no	ECE	QA	none
<i>Likelihood and logit read-off</i>									
Ott et al. [2018]	2018	score	white	yes	no	no	other	generation	none
Lovering et al. [2025]	2024	typed	white	no	no	no	other	reasoning	none
Plaut et al. [2024]	2024	typed	white	no	no	no	multiple	QA	abstain
Spieß et al. [2024]	2024	score	white	no	no	no	multiple	code	none
Wang et al. [2024d]	2024	typed	white	no	no	no	none	QA	none
<i>Self-evaluation and read-off calibration</i>									
Kadavath et al. [2022]	2022	score	white	no	no	partial	multiple	QA	none

Table F1, continued

Work	Yr	Contract	Access	Samples	Guar.	Trained	Metric	Task	Action
S1 post-hoc map (52)									
<i>Fixed maps over the read-off</i>									
Zhang et al. [2023c]	2023	score	white	no	no	no	other	generation	none
Bakman et al. [2024]	2024	score	white	no	no	partial	AUROC	QA	none
Fadeeva et al. [2024]	2024	score	white	no	no	no	AUROC	generation	none
Li et al. [2025b]	2025	score	white	no	no	no	multiple	reasoning	none
Vazhentsev et al. [2026]	2025	score	white	no	no	no	AUROC	generation	none
Badhe et al. [2026]	2026	typed	white	no	no	no	multiple	classification	none
<i>Label-bias and option-prior debiasing</i>									
Zheng et al. [2023]	2023	typed	white	no	no	no	none	QA	none
Reif and Schwartz [2024]	2024	typed	white	no	no	no	none	classification	none
<i>Learned calibrators</i>									
Ye and Durrett [2022]	2021	score	black	no	no	partial	multiple	QA	none
Zhang et al. [2021]	2021	score	white	no	no	partial	selective	QA	filter
Dhuliawala et al. [2022]	2022	score	white	no	no	partial	multiple	QA	none
Han et al. [2022]	2022	typed	white	no	no	partial	none	classification	none
Liu et al. [2024a]	2024	score	white	no	no	partial	multiple	QA	none
Liu et al. [2024d]	2024	score	either	no	no	partial	AUROC	QA	none
Ulmer et al. [2024a]	2024	score	black	no	no	partial	multiple	QA	none
Radharapu et al. [2025]	2025	score	white	no	no	partial	ECE	judging	none
Luo et al. [2026]	2026	typed	white	no	no	no	multiple	QA	none
<i>Multicalibration and task-specific recalibration</i>									
Detommaso et al. [2024]	2024	score	either	no	no	no	multiple	QA	none
Liu et al. [2024b]	2024	typed	white	no	no	no	ECE	classification	none
<i>Probes and learned confidence heads</i>									
Kumar and Sarawagi [2019]	2019	score	white	no	no	partial	ECE	generation	none
Mielke et al. [2022]	2020	free-text	either	no	no	partial	multiple	QA	none
Burns et al. [2022]	2022	typed	white	no	no	partial	other	classification	none
Azaria and Mitchell [2023]	2023	typed	white	no	no	partial	AUROC	classification	none
CH-Wang et al. [2024]	2023	score	white	no	no	partial	other	generation	none
Liu et al. [2023a]	2023	typed	white	no	no	partial	ECE	classification	none
Liu et al. [2023c]	2023	score	white	no	no	partial	multiple	generation	none
Slobodkin et al. [2023]	2023	free-text	white	no	no	partial	other	QA	none
Beigi et al. [2024a]	2024	score	white	no	no	partial	ECE	QA	none
Du et al. [2024]	2024	score	white	no	no	partial	AUROC	QA	none
Gottesman and Geva [2024]	2024	score	white	no	no	partial	other	QA	none
Ji et al. [2024]	2024	score	white	no	no	partial	other	generation	none
Kossen et al. [2024]	2024	score	white	no	no	partial	AUROC	QA	none
Orgad et al. [2024]	2024	score	white	no	no	partial	AUROC	QA	none
Su et al. [2024b]	2024	score	white	no	no	partial	AUROC	generation	none
Khanmohammadi et al. [2025]	2025	score	white	no	no	partial	multiple	QA	none
Lu [2026]	2026	score	white	no	no	partial	AUROC	reasoning	none
<i>Prompt and context calibration</i>									
Holtzman et al. [2021]	2021	typed	white	no	no	no	none	classification	none
Zhao et al. [2021]	2021	typed	white	no	no	no	none	classification	none
Fei et al. [2023]	2023	typed	white	no	no	no	none	classification	none
Jiang et al. [2023b]	2023	typed	white	yes	no	no	none	classification	none
Liusie et al. [2023]	2023	typed	white	no	no	no	none	classification	none
Zhou et al. [2023b]	2023	typed	white	no	no	no	none	classification	none
Abbas et al. [2024]	2024	typed	white	no	no	no	ECE	classification	none
Cho et al. [2025]	2024	typed	white	no	no	no	none	classification	none
Gundem et al. [2025]	2025	typed	white	no	no	no	none	classification	none
<i>Scaling and binning</i>									
Guo et al. [2017]	2017	typed	white	no	no	no	multiple	classification	none
Kull et al. [2019]	2019	typed	white	no	no	no	multiple	classification	none
Kumar et al. [2019]	2019	typed	either	no	yes	no	multiple	classification	none
Gupta and Ramdas [2021]	2021	typed	either	no	yes	no	ECE	classification	none
Shen et al. [2024]	2024	typed	white	no	no	partial	multiple	QA	none
Xie et al. [2024a]	2024	score	white	no	no	partial	multiple	QA	none
Luo et al. [2025]	2025	typed	white	no	no	no	multiple	QA	none
S2 prompt elicitation (54)									
<i>Abstention prompting</i>									
Zong et al. [2026]	2026	score	black	no	no	no	multiple	QA	abstain
<i>Introspection, deliberation and task-specific elicitation</i>									
Sun et al. [2024]	2024	constrained	either	no	no	partial	multiple	other	none
Yang et al. [2024c]	2024	score	black	yes	no	no	multiple	QA	none
Mei et al. [2025]	2025	score	black	no	no	no	ECE	reasoning	none
<i>Linguistic hedges and epistemic markers</i>									
Sileo and Moens [2022]	2022	free-text	white	no	no	no	none	reasoning	none

Table F1, continued

Work	Yr	Contract	Access	Samples	Guar.	Trained	Metric	Task	Action
Ren et al. [2023b]	2023	score	white	yes	no	no	multiple	QA	none
Singh et al. [2023]	2023	score	black	no	no	no	other	other	none
Yin et al. [2023]	2023	free-text	black	no	no	no	other	QA	none
Ghafouri et al. [2024]	2024	free-text	either	no	no	partial	other	QA	none
Huang et al. [2024b]	2024	score	black	yes	no	no	multiple	generation	hand off
Kumar et al. [2024]	2024	score	white	no	no	no	other	QA	none
Li et al. [2024a]	2024	free-text	black	no	no	no	other	reasoning	filter
Li et al. [2024e]	2024	score	either	?	no	no	multiple	QA	none
Pawitan and Holmes [2024]	2024	score	black	no	no	no	ECE	reasoning	none
Sicilia et al. [2024]	2024	score	either	no	no	partial	Brier	QA	none
Tang et al. [2024]	2024	free-text	black	no	no	no	other	other	none
Xue et al. [2024]	2024	score	either	?	no	no	multiple	QA	none
Zhao et al. [2024b]	2024	score	black	no	no	no	ECE	QA	retrieve
Liu et al. [2025e]	2025	free-text	black	no	no	no	ECE	QA	none
Sun et al. [2025]	2025	score	black	no	no	no	ECE	reasoning	none
Liu and Cohan [2026]	2026	free-text	black	no	no	no	other	QA	none
<i>Self-evaluation prompts</i>									
Tian et al. [2025]	2025	score	black	no	no	no	multiple	judging	none
<i>Stated confidence that triggers an action</i>									
Ni et al. [2024a]	2024	free-text	black	no	no	no	other	QA	retrieve
Ferrer et al. [2026]	2026	score	black	no	no	no	multiple	judging	hand off
<i>Verbalised numeric confidence</i>									
Tanneru et al. [2023]	2023	score	black	yes	no	no	other	QA	none
Tian et al. [2023]	2023	score	black	no	no	no	multiple	QA	none
Xiong et al. [2023]	2023	score	black	yes	no	no	multiple	reasoning	none
Zhou et al. [2023a]	2023	free-text	black	no	no	no	ECE	QA	none
Belem et al. [2024]	2024	free-text	black	no	no	no	other	other	none
Groot and Valdenegro-Toro [2024]	2024	score	black	no	no	no	ECE	other	none
Lee et al. [2024a]	2024	free-text	black	no	no	no	none	judging	none
Ni et al. [2024b]	2024	score	either	no	no	no	other	QA	none
Rivera et al. [2024]	2024	score	black	yes	no	no	multiple	classification	none
Wang et al. [2024f]	2024	score	black	no	no	no	multiple	classification	none
Wei et al. [2024]	2024	score	black	yes	no	no	ECE	QA	none
Yang et al. [2024d]	2024	score	black	no	no	no	ECE	QA	none
Yona et al. [2024]	2024	free-text	black	yes	no	no	other	QA	none
Chhikara [2025]	2025	score	black	no	no	no	ECE	QA	none
Liu et al. [2025d]	2025	free-text	black	yes	no	no	other	QA	none
Naderi et al. [2025]	2025	score	black	no	no	no	multiple	QA	none
Obadinma and Zhu [2025]	2025	score	black	no	no	no	other	QA	none
Tao et al. [2025b]	2025	free-text	black	no	no	partial	multiple	QA	none
Tao et al. [2025a]	2025	score	either	no	no	no	multiple	QA	none
Wang and Stengel-Eskin [2025]	2025	score	black	yes	no	no	multiple	QA	none
Wang et al. [2025d]	2025	score	black	no	no	no	multiple	QA	none
Yoon et al. [2025]	2025	score	black	no	no	no	multiple	reasoning	none
Yu et al. [2025b]	2025	score	black	yes	no	no	multiple	agentic	none
Zhou et al. [2025]	2025	score	black	yes	no	no	multiple	QA	none
Cacioli [2026]	2026	score	black	no	no	no	AUROC	QA	none
ffrench Constant et al. [2026]	2026	score	black	no	no	no	Brier	QA	none
Hsiao [2026]	2026	score	black	no	no	no	ECE	judging	none
Kumaran et al. [2026]	2026	score	white	no	no	no	multiple	QA	none
Marashian et al. [2026]	2026	score	black	no	no	no	ECE	generation	none
Yang et al. [2026a]	2026	score	black	no	no	no	multiple	QA	none
S3 extra inference (121)									
<i>Benchmarks and empirical studies</i>									
Fadeeva et al. [2023]	2023	free-text	either	?	no	no	AUROC	generation	none
Huang et al. [2025c]	2023	free-text	either	yes	no	no	AUROC	other	none
Huang et al. [2024a]	2024	free-text	either	yes	no	no	AUROC	QA	none
Ling et al. [2024]	2024	typed	white	yes	no	no	AUROC	classification	none
Vashurin et al. [2025]	2024	free-text	either	?	no	no	multiple	generation	none
<i>Conformal abstention, selection and cascades</i>									
Fisch et al. [2020]	2020	set	white	no	yes	no	coverage	QA	set
Lee et al. [2024b]	2023	free-text	either	no	yes	partial	coverage	generation	abstain
Yadkori et al. [2024b]	2024	free-text	black	yes	yes	no	coverage	QA	abstain
Gui et al. [2024]	2024	free-text	either	yes	yes	partial	coverage	generation	abstain
Jung et al. [2024]	2024	typed	either	yes	yes	no	multiple	judging	hand off
Tayebati et al. [2025]	2025	set	white	no	no	partial	multiple	QA	abstain
<i>Conformal factuality and claim filtering</i>									
Cherian et al. [2024]	2024	free-text	either	yes	yes	partial	coverage	generation	filter
Mohri and Hashimoto [2024]	2024	free-text	either	yes	yes	no	coverage	generation	filter
Feng et al. [2025]	2025	free-text	black	yes	yes	no	coverage	QA	filter

Table F1, continued

Work	Yr	Contract	Access	Samples	Guar.	Trained	Metric	Task	Action
Jiang et al. [2025]	2025	free-text	black	yes	yes	partial	coverage	QA	filter
Lee et al. [2025]	2025	free-text	either	?	yes	no	coverage	QA	abstain
Rubin-Toles et al. [2025]	2025	free-text	black	yes	yes	no	coverage	reasoning	filter
<i>Conformal prediction sets</i>									
Dey et al. [2022]	2021	set	white	no	yes	no	coverage	classification	set
Fisch et al. [2021]	2021	set	white	no	yes	partial	coverage	classification	set
Choubey et al. [2022]	2022	set	white	no	yes	no	coverage	classification	set
Deuschmann et al. [2023]	2023	set	white	no	yes	no	coverage	generation	set
Kumar et al. [2023]	2023	set	white	no	yes	no	coverage	QA	set
Li et al. [2024d]	2023	set	either	yes	yes	no	coverage	QA	set
Quach et al. [2023]	2023	set	either	yes	yes	no	coverage	generation	set
Ravfogel et al. [2023]	2023	set	white	no	yes	no	coverage	generation	set
Ren et al. [2023a]	2023	set	white	no	yes	no	coverage	robotics	hand off
Wang et al. [2025f]	2023	set	white	no	yes	no	coverage	robotics	hand off
Zerva and Martins [2024]	2023	set	white	?	yes	no	coverage	judging	set
Kaur et al. [2024]	2024	set	black	yes	yes	no	multiple	QA	set
Kostumov et al. [2024]	2024	set	white	no	yes	no	multiple	QA	set
Liang et al. [2024b]	2024	set	white	no	yes	no	coverage	robotics	hand off
Liu and Wu [2024]	2024	free-text	black	yes	yes	partial	multiple	generation	filter
Ren et al. [2024]	2024	set	white	no	yes	no	other	robotics	adapt compute
Su et al. [2024a]	2024	set	black	yes	yes	no	coverage	QA	set
Ulmer et al. [2024b]	2024	set	white	no	yes	no	coverage	generation	set
Vishwakarma et al. [2024]	2024	set	white	no	yes	partial	coverage	QA	set
Wang et al. [2024a]	2024	set	black	yes	yes	no	multiple	QA	set
Wang et al. [2025c]	2024	set	white	no	yes	no	coverage	robotics	hand off
Ye et al. [2024]	2024	set	white	no	yes	no	multiple	QA	set
Zhang et al. [2024b]	2024	set	either	no	yes	no	coverage	QA	set
Chan et al. [2025]	2025	set	?	?	yes	no	coverage	QA	adapt compute
Doula et al. [2025]	2025	set	black	yes	yes	no	coverage	robotics	hand off
Huang et al. [2025b]	2025	set	white	no	yes	no	multiple	QA	hand off
Kaur et al. [2025]	2025	set	black	yes	yes	no	none	reasoning	set
Kuwahara et al. [2025]	2025	free-text	either	no	yes	no	coverage	generation	filter
Noorani et al. [2025]	2025	set	black	yes	yes	no	coverage	generation	set
Shahrokhi et al. [2025]	2025	set	black	yes	yes	no	coverage	code	set
Sheng et al. [2025]	2025	set	either	no	yes	no	coverage	judging	set
Sundarsingh et al. [2025]	2025	set	?	?	yes	no	coverage	robotics	hand off
Wang et al. [2025g]	2025	set	black	yes	yes	no	coverage	QA	abstain
Yang and Liu [2025]	2025	set	black	yes	yes	no	multiple	QA	set
Emmenegger et al. [2026]	2026	free-text	?	yes	yes	no	coverage	generation	abstain
Zhang et al. [2026b]	2026	set	?	?	yes	no	none	robotics	adapt compute
<i>Ensembles and cross-model disagreement</i>									
Dong et al. [2018]	2018	constrained	white	yes	no	partial	selective	code	none
Hou et al. [2024]	2023	free-text	black	yes	no	no	AUROC	QA	hand off
Tonolini et al. [2024]	2024	typed	black	yes	no	no	multiple	classification	none
Kruse et al. [2025]	2025	typed	?	yes	no	no	multiple	classification	none
Hamidieh et al. [2026]	2026	free-text	black	yes	no	no	multiple	generation	none
<i>Internal-state signals</i>									
Fomicheva et al. [2020]	2020	score	white	yes	no	no	other	generation	none
Chen et al. [2024]	2024	free-text	white	yes	no	no	AUROC	QA	none
Li et al. [2026e]	2025	free-text	black	yes	no	no	AUROC	QA	none
<i>Long-form and claim-level</i>									
Da et al. [2024]	2024	free-text	black	yes	no	no	multiple	QA	none
Jiang et al. [2024]	2024	free-text	black	yes	no	no	AUROC	generation	filter
Zhang et al. [2025d]	2024	free-text	either	yes	no	no	multiple	generation	filter
Zhang et al. [2024c]	2024	free-text	black	yes	no	no	selective	generation	filter
Fan et al. [2026]	2026	free-text	black	yes	no	no	AUROC	generation	none
<i>Risk control and early exit</i>									
Angelopoulos et al. [2025]	2021	set	either	no	yes	no	coverage	other	set
Schuster et al. [2021]	2021	typed	white	no	yes	partial	coverage	classification	adapt compute
Angelopoulos et al. [2022]	2022	set	either	no	yes	no	coverage	other	set
Schuster et al. [2022]	2022	score	white	no	yes	partial	other	generation	adapt compute
Zollo et al. [2023]	2023	free-text	black	no	yes	no	coverage	generation	none
Jazbec et al. [2024]	2024	typed	white	no	yes	no	multiple	other	adapt compute
Kang et al. [2024]	2024	set	black	yes	yes	no	coverage	generation	set
Kladny et al. [2024]	2024	set	black	yes	yes	no	coverage	generation	set
Wang et al. [2024b]	2024	set	black	yes	yes	no	coverage	QA	set
Chen et al. [2025b]	2025	free-text	black	yes	yes	no	coverage	generation	filter
Ke et al. [2025]	2025	set	black	yes	yes	no	multiple	QA	set
Overman and Bayati [2025]	2025	score	black	no	yes	no	coverage	QA	hand off
Wang et al. [2025b]	2025	set	black	yes	yes	no	coverage	QA	abstain

Table F1, continued

Work	Yr	Contract	Access	Samples	Guar.	Trained	Metric	Task	Action
Ye [2025]	2025	set	black	yes	yes	no	coverage	QA	set
Feng et al. [2026]	2026	constrained	either	?	yes	no	other	agentic	hand off
Wang et al. [2026b]	2026	score	white	no	yes	no	coverage	reasoning	abstain
<i>Sampling consistency</i>									
Wang et al. [2023b]	2022	free-text	black	yes	no	no	other	reasoning	none
Agrawal et al. [2023]	2023	free-text	black	yes	no	no	multiple	other	none
Chen and Mueller [2023]	2023	free-text	black	yes	no	no	AUROC	QA	filter
Cohen et al. [2023]	2023	free-text	black	yes	no	no	other	QA	none
Cole et al. [2023]	2023	free-text	black	yes	no	no	multiple	QA	abstain
Friel and Sanyal [2023]	2023	free-text	black	yes	no	no	AUROC	judging	none
Liu et al. [2023b]	2023	typed	either	yes	no	no	other	judging	none
Manakul et al. [2023]	2023	free-text	black	yes	no	no	other	generation	none
Mündler et al. [2023]	2023	free-text	black	yes	no	no	other	generation	filter
Portillo Wightman et al. [2023]	2023	typed	either	yes	no	no	ECE	QA	none
Zhao et al. [2024a]	2023	free-text	black	yes	no	no	other	QA	none
Becker and Soatto [2024]	2024	typed	white	yes	no	no	multiple	QA	none
Gao et al. [2024]	2024	free-text	black	yes	no	no	ECE	QA	none
Lyu et al. [2025]	2024	free-text	black	yes	no	no	multiple	reasoning	none
Yadkori et al. [2024a]	2024	free-text	black	yes	no	no	selective	QA	abstain
Yang et al. [2024a]	2024	typed	black	yes	no	no	multiple	QA	none
Zhang et al. [2024a]	2024	typed	either	yes	no	no	ECE	QA	none
Tan et al. [2025]	2025	free-text	white	yes	no	partial	AUROC	QA	none
<i>Semantic entropy and meaning clustering</i>									
Kuhn et al. [2023]	2023	free-text	white	yes	no	no	AUROC	QA	none
Lin et al. [2024]	2023	free-text	black	yes	no	no	multiple	QA	none
Zhang et al. [2023b]	2023	free-text	black	yes	no	no	AUROC	QA	none
Aichberger et al. [2025]	2024	free-text	white	yes	no	no	AUROC	QA	none
Ao et al. [2024]	2024	free-text	black	yes	no	no	multiple	QA	none
Farquhar et al. [2024]	2024	free-text	either	yes	no	no	multiple	QA	none
Grewal et al. [2024]	2024	free-text	either	yes	no	partial	AUROC	QA	none
Nikitin et al. [2024]	2024	free-text	either	yes	no	no	multiple	QA	none
Qiu and Mäkeläinen [2024]	2024	free-text	white	yes	no	no	AUROC	QA	none
Zhang et al. [2024e]	2024	free-text	black	yes	no	no	other	QA	none
Ciosek et al. [2025]	2025	free-text	?	yes	no	no	AUROC	QA	adapt compute
Ma et al. [2025]	2025	free-text	white	yes	no	no	AUROC	QA	none
Nguyen et al. [2025]	2025	free-text	black	yes	no	no	AUROC	generation	none
Sharma and David [2025]	2025	free-text	?	yes	no	no	other	code	abstain
Zhao et al. [2026]	2025	free-text	black	yes	no	no	multiple	QA	none
Bouchard et al. [2026]	2026	free-text	black	yes	no	no	AUROC	code	none
He et al. [2026a]	2026	free-text	black	yes	no	no	AUROC	code	abstain
Sun et al. [2026a]	2026	free-text	?	yes	no	no	AUROC	QA	adapt compute
<i>Token-level aggregation</i>									
Malinin and Gales [2020]	2020	free-text	white	yes	no	no	AUROC	generation	none
Duan et al. [2024]	2023	free-text	white	yes	no	no	AUROC	QA	none
Wang et al. [2025a]	2024	free-text	white	yes	no	no	AUROC	QA	filter
S4 training (80): listed in Table C1									
Theory and analysis (5): listed in Table C1									
X consumers and contract (71): output-contract works detailed in Table D1									
<i>Abstention</i>									
Kamath et al. [2020]	2020	score	white	no	no	partial	selective	QA	abstain
Feng et al. [2024]	2024	free-text	black	no	no	no	ECE	QA	abstain
Srinivasan et al. [2024]	2024	score	white	no	no	no	multiple	QA	adapt compute
<i>Adaptive compute</i>									
Aggarwal et al. [2023]	2023	free-text	black	yes	no	no	none	reasoning	adapt compute
Jiang et al. [2023a]	2023	score	white	no	no	no	none	QA	retrieve
Damani et al. [2024]	2024	score	white	yes	no	partial	multiple	reasoning	hand off
Li et al. [2024f]	2024	free-text	black	yes	no	no	none	reasoning	adapt compute
Fu et al. [2025]	2025	score	white	yes	no	no	other	reasoning	filter
Yang et al. [2025b]	2025	score	white	no	no	no	other	reasoning	adapt compute
<i>Confidence-weighted selection</i>									
Taubenfeld et al. [2025]	2025	score	either	yes	no	no	multiple	reasoning	filter
<i>Constrained decoding</i>									
Scholak et al. [2021]	2021	constrained	white	no	no	no	none	code	none
Beurer-Kellner et al. [2023]	2022	constrained	white	no	no	no	none	generation	none
Poesia et al. [2022]	2022	constrained	white	no	no	partial	none	code	none
Geng et al. [2023]	2023	constrained	white	no	no	no	none	other	none
Lew et al. [2023]	2023	constrained	white	yes	no	no	none	generation	none
Willard and Louf [2023]	2023	constrained	white	no	no	no	none	generation	none

Table F1, continued

Work	Yr	Contract	Access	Samples	Guar.	Trained	Metric	Task	Action
Beurer-Kellner et al. [2024]	2024	constrained	white	no	no	no	none	generation	none
Dong et al. [2025]	2024	constrained	white	no	no	no	none	generation	none
Geng et al. [2024b]	2024	constrained	black	no	no	no	none	other	none
Koo et al. [2024]	2024	constrained	white	no	no	no	none	generation	none
Park et al. [2024]	2024	constrained	white	yes	no	no	none	code	none
Ugare et al. [2024]	2024	constrained	white	no	no	no	none	code	none
Banerjee et al. [2025]	2025	constrained	white	no	no	no	none	reasoning	none
Li et al. [2026b]	2025	constrained	white	no	no	no	none	code	none
Loula et al. [2025]	2025	constrained	white	yes	no	no	other	code	none
Mündler et al. [2025]	2025	constrained	white	no	no	no	none	code	none
Park et al. [2025b]	2025	constrained	white	no	no	no	none	generation	none
Wang et al. [2025e]	2025	free-text	either	no	no	partial	none	generation	none
Reddy et al. [2026]	2026	constrained	white	no	no	no	none	reasoning	none
<i>Deferral</i>									
Jitkrittum et al. [2023]	2023	typed	white	no	no	partial	none	classification	hand off
Fanconi and van der Schaar [2025]	2025	score	either	?	no	partial	none	QA	hand off
Wu et al. [2025b]	2025	score	black	?	no	no	other	reasoning	hand off
Kondadadi and Ortega [2026]	2026	typed	either	?	no	partial	none	classification	hand off
<i>Human reliance on model output and stated confidence</i>									
Si et al. [2024]	2023	free-text	black	no	no	no	other	QA	none
Kim et al. [2024]	2024	free-text	black	no	no	no	other	QA	none
Steyvers et al. [2025b]	2024	free-text	either	no	no	no	multiple	QA	none
Zhou et al. [2024b]	2024	free-text	black	no	no	no	other	QA	none
Zhou et al. [2024a]	2024	free-text	black	no	no	no	other	QA	none
Rathi et al. [2025]	2025	free-text	black	no	no	no	other	QA	none
Villavicencio et al. [2026]	2026	score	either	?	no	no	other	QA	none
<i>Judge calibration</i>									
Liu et al. [2023d]	2023	score	black	no	no	no	none	judging	none
Wang et al. [2023a]	2023	free-text	black	yes	no	no	none	judging	hand off
Lee et al. [2026b]	2025	free-text	black	no	no	no	coverage	judging	none
Li et al. [2026c]	2026	typed	black	no	no	no	multiple	judging	hand off
<i>Routing/cascade</i>									
Varshney and Baral [2022]	2022	typed	white	no	no	no	none	classification	hand off
Aggarwal et al. [2024]	2023	score	black	yes	no	partial	none	QA	hand off
Chen et al. [2023b]	2023	score	black	no	no	partial	none	QA	hand off
Yue et al. [2024]	2023	free-text	black	yes	no	no	other	reasoning	hand off
Ding et al. [2024]	2024	score	black	no	no	partial	none	generation	hand off
Gupta et al. [2024]	2024	score	white	no	no	partial	none	generation	hand off
Hu et al. [2024]	2024	score	black	no	no	partial	none	QA	hand off
Narasimhan et al. [2024]	2024	score	white	no	no	no	none	generation	hand off
Ong et al. [2024]	2024	score	black	no	no	partial	none	generation	hand off
Ramírez et al. [2024]	2024	typed	white	no	no	no	none	classification	hand off
Soiffer et al. [2025]	2025	free-text	black	yes	no	no	none	generation	hand off
Zellinger and Thomson [2025]	2025	typed	white	no	no	no	ECE	QA	hand off
Bouchard [2026]	2026	score	either	?	no	no	other	QA	hand off
<i>Structured-output benchmark</i>									
Tang et al. [2023]	2023	constrained	either	no	no	no	none	generation	none
Shorten et al. [2024]	2024	constrained	black	no	no	no	none	QA	none
Tam et al. [2024]	2024	constrained	either	no	no	no	none	reasoning	none
Xia et al. [2024a]	2024	constrained	black	no	no	no	none	generation	none
Geng et al. [2025]	2025	constrained	white	no	no	no	none	generation	none
Yang et al. [2025a]	2025	constrained	black	no	no	no	none	generation	none
Fan [2026]	2026	constrained	black	no	no	no	none	reasoning	none
Galeone et al. [2026]	2026	constrained	black	no	no	no	none	reasoning	none
Lee et al. [2026a]	2026	constrained	either	no	no	no	none	reasoning	none
Singh et al. [2026a]	2026	constrained	black	no	no	no	none	QA	none
<i>Studies and benchmarks of abstention</i>									
Tomani et al. [2024]	2024	free-text	either	yes	no	no	multiple	QA	none
Kirichenko et al. [2025]	2025	free-text	black	no	no	no	other	reasoning	none
Wang et al. [2026c]	2026	score	black	no	no	no	multiple	QA	abstain
<i>Typed classification</i>									
Wang et al. [2024c]	2024	typed	white	no	no	no	none	classification	none